paul graham

# on money and investors

essays, volume six

october 2026

a printed book of essays

paul graham

# on money and investors

essays, volume six

printed at home, october 2026

12 essays, 37,632 words. all of them are at paulgraham.com, free, as they should be. this is one reader's copy, printed at home to be read away from a screen, with a pencil.

a printed book of essays
github.com/workinprogmess/a-printed-book-of-essays

# contents


| | | | |
|---|---|---|---|
| 01 | how to make wealth | | 4 |
| 02 | how people get rich now | april 2021 | 20 |
| 03 | billionaires build | december 2020 | 26 |
| 04 | ramen profitable | | 34 |
| 05 | default alive or default dead? | october 2015 | 38 |
| 06 | the fatal pinch | december 2014 | 42 |
| 07 | the equity equation | july 2007 | 46 |
| 08 | investor herd dynamics | | 50 |
| 09 | high resolution fundraising | | 54 |
| 10 | how to convince investors | | 58 |
| 11 | how to raise money | | 66 |
| 12 | donate unrestricted | march 2021 | 84 |

01

# how to make wealth

paulgraham.com/wealth.html

may 2004

*(this essay was originally published in hackers & painters.)*

if you wanted to get rich, how would you do it? i think your best bet would be to start or join a startup. that's been a reliable way to get rich for hundreds of years. the word "startup" dates from the 1960s, but what happens in one is very similar to the venture-backed trading voyages of the middle ages.

startups usually involve technology, so much so that the phrase "high-tech startup" is almost redundant. a startup is a small company that takes on a hard technical problem.

lots of people get rich knowing nothing more than that. you don't have to know physics to be a good pitcher. but i think it could give you an edge to understand the underlying principles. why do startups have to be small? will a startup inevitably stop being a startup as it grows larger? and why do they so often work on developing new technology? why are there so many startups selling new drugs or computer software, and none selling corn oil or laundry detergent?

**the proposition**

economically, you can think of a startup as a way to compress your whole working life into a few years. instead of working at a low intensity for forty years, you work as hard as you possibly can for four. this pays especially well in technology, where you earn a premium for working fast.

here is a brief sketch of the economic proposition. if you're a good hacker in your mid twenties, you can get a job paying about $80,000 per year. so on average such a hacker must be able to do at least $80,000 worth of work per year for the company just to break even. you could probably work twice as many hours as a corporate employee, and if you focus you can probably get three times as much done in an hour.[1] you should get another multiple of two, at least, by eliminating the drag of the pointy-haired middle manager who would be your boss in a big company. then there is one more multiple: how much smarter are you than your job description expects you to be? suppose another multiple of three. combine all these multipliers, and i'm claiming you could be 36 times more productive than you're expected to be in a random corporate job.[2] if a fairly good hacker is worth $80,000 a year at a big company, then a smart hacker working very hard without any corporate bullshit to slow him down should be able to do work worth about $3 million a year.

like all back-of-the-envelope calculations, this one has a lot of wiggle room. i wouldn't try to defend the actual numbers. but i stand by the structure of the calculation. i'm not claiming the multiplier is precisely 36, but it is certainly more than 10, and probably rarely as high as 100.

if $3 million a year seems high, remember that we're talking about the limit case: the case where you not only have zero leisure time but indeed work so hard that you endanger your health.

startups are not magic. they don't change the laws of wealth creation. they just represent a point at the far end of the curve. there is a conservation law at work here: if you want to make a million dollars, you have to endure a million dollars' worth of pain. for example, one way to make a million dollars would be to work for the post office your whole life, and save every penny of your salary. imagine the stress of working for the post office for fifty years. in a startup you compress all this stress into three or four years. you do tend to get a certain bulk discount if you buy the economy-size pain, but you can't evade the fundamental conservation law. if starting a startup were easy, everyone would do it.

**millions, not billions**

if $3 million a year seems high to some people, it will seem low to others. three *million?* how do i get to be a billionaire, like bill gates?

so let's get bill gates out of the way right now. it's not a good idea to use famous rich people as examples, because the press only write about the very richest, and these tend to be outliers. bill gates is a smart, determined, and hardworking man, but you need more than that to make as much money as he has. you also need to be very lucky.

there is a large random factor in the success of any company. so the guys you end up reading about in the papers are the ones who are very smart, totally dedicated, *and* win the lottery. certainly bill is smart and dedicated, but microsoft also happens to have been the beneficiary of one of the most spectacular blunders in the history of business: the licensing deal for dos. no doubt bill did everything he could to steer ibm into making that blunder, and he has done an excellent job of exploiting it, but if there had been one person with a brain on ibm's side, microsoft's future would have been very different. microsoft at that stage had little leverage over ibm. they were effectively a component supplier. if ibm had required an exclusive license, as they should have, microsoft would still have signed the deal. it would still have meant a lot of money for them, and ibm could easily have gotten an operating system elsewhere.

instead ibm ended up using all its power in the market to give microsoft control of the pc standard. from that point, all microsoft had to do was execute. they never had to bet the company on a bold decision. all they had to do was play hardball with licensees and copy more innovative products reasonably promptly.

if ibm hadn't made this mistake, microsoft would still have been a successful company, but it could not have grown so big so fast. bill gates would be rich, but he'd be somewhere near the bottom of the forbes 400 with the other guys his age.

there are a lot of ways to get rich, and this essay is about only one of them. this essay is about how to make money by creating wealth and getting paid for it. there are plenty of other ways to get money, including chance, speculation, marriage, inheritance, theft, extortion, fraud, monopoly, graft, lobbying, counterfeiting, and prospecting. most of the greatest fortunes have probably involved several of these.

the advantage of creating wealth, as a way to get rich, is not just that it's more legitimate (many of the other methods are now illegal) but that it's more *straightforward.* you just have to do something people want.

**money is not wealth**

if you want to create wealth, it will help to understand what it is. wealth is not the same thing as money.[3] wealth is as old as human history. far older, in fact; ants have wealth. money is a comparatively recent invention.

wealth is the fundamental thing. wealth is stuff we want: food, clothes, houses, cars, gadgets, travel to interesting places, and so on. you can have wealth without having money. if you had a magic machine that could on command make you a car or cook you dinner or do your laundry, or do anything else you wanted, you wouldn't need money. whereas if you were in the middle of antarctica, where there is nothing to buy, it wouldn't matter how much money you had.

wealth is what you want, not money. but if wealth is the important thing, why does everyone talk about making money? it is a kind of shorthand: money is a way of moving wealth, and in practice they

are usually interchangeable. but they are not the same thing, and unless you plan to get rich by counterfeiting, talking about *making money* can make it harder to understand how to make money.

money is a side effect of specialization. in a specialized society, most of the things you need, you can't make for yourself. if you want a potato or a pencil or a place to live, you have to get it from someone else.

how do you get the person who grows the potatoes to give you some? by giving him something he wants in return. but you can't get very far by trading things directly with the people who need them. if you make violins, and none of the local farmers wants one, how will you eat?

the solution societies find, as they get more specialized, is to make the trade into a two-step process. instead of trading violins directly for potatoes, you trade violins for, say, silver, which you can then trade again for anything else you need. the intermediate stuff-- the *medium of exchange*-- can be anything that's rare and portable. historically metals have been the most common, but recently we've been using a medium of exchange, called the *dollar*, that doesn't physically exist. it works as a medium of exchange, however, because its rarity is guaranteed by the u.s. government.

the advantage of a medium of exchange is that it makes trade work. the disadvantage is that it tends to obscure what trade really means. people think that what a business does is make money. but money is just the intermediate stage-- just a shorthand-- for whatever people want. what most businesses really do is make wealth. they do something people want.[4]

**the pie fallacy**

a surprising number of people retain from childhood the idea that there is a fixed amount of wealth in the world. there is, in any normal family, a fixed amount of *money* at any moment. but that's not the same thing.

when wealth is talked about in this context, it is often described as a pie. "you can't make the pie larger," say politicians. when you're talking about the amount of money in one family's bank account, or the amount available to a government from one year's tax revenue, this is true. if one person gets more, someone else has to get less.

i can remember believing, as a child, that if a few rich people had all the money, it left less for everyone else. many people seem to continue to believe something like this well into adulthood. this fallacy is usually there in the background when you hear someone talking about how x percent of the population have y percent of the wealth. if you plan to start a startup, then whether you realize it or not, you're planning to disprove the pie fallacy.

what leads people astray here is the abstraction of money. money is not wealth. it's just something we use to move wealth around. so although there may be, in certain specific moments (like your family, this month) a fixed amount of money available to trade with other people for things you want, there is not a fixed amount of wealth in the world. *you can make more wealth.* wealth has been getting created and destroyed (but on balance, created) for all of human history.

suppose you own a beat-up old car. instead of sitting on your butt next summer, you could spend the time restoring your car to pristine condition. in doing so you create wealth. the world is-- and you specifically are-- one pristine old car the richer. and not just in some metaphorical way. if you sell your car, you'll get more for it.

in restoring your old car you have made yourself richer. you haven't made anyone else poorer. so there is obviously not a fixed pie. and in fact, when you look at it this way, you wonder why anyone would

think there was.[5]

kids know, without knowing they know, that they can create wealth. if you need to give someone a present and don't have any money, you make one. but kids are so bad at making things that they consider home-made presents to be a distinct, inferior, sort of thing to store-bought ones-- a mere expression of the proverbial thought that counts. and indeed, the lumpy ashtrays we made for our parents did not have much of a resale market.

**craftsmen**

the people most likely to grasp that wealth can be created are the ones who are good at making things, the craftsmen. their hand-made objects become store-bought ones. but with the rise of industrialization there are fewer and fewer craftsmen. one of the biggest remaining groups is computer programmers.

a programmer can sit down in front of a computer and *create wealth*. a good piece of software is, in itself, a valuable thing. there is no manufacturing to confuse the issue. those characters you type are a complete, finished product. if someone sat down and wrote a web browser that didn't suck (a fine idea, by the way), the world would be that much richer. [5b]

everyone in a company works together to create wealth, in the sense of making more things people want. many of the employees (e.g. the people in the mailroom or the personnel department) work at one remove from the actual making of stuff. not the programmers. they literally think the product, one line at a time. and so it's clearer to programmers that wealth is something that's made, rather than being distributed, like slices of a pie, by some imaginary daddy.

it's also obvious to programmers that there are huge variations in the rate at which wealth is created. at viaweb we had one programmer who was a sort of monster of productivity. i remember watching what he did one long day and estimating that he had added several hundred thousand dollars to the market value of the company. a great programmer, on a roll, could create a million dollars worth of wealth in a couple weeks. a mediocre programmer over the same period will generate zero or even negative wealth (e.g. by introducing bugs).

this is why so many of the best programmers are libertarians. in our world, you sink or swim, and there are no excuses. when those far removed from the creation of wealth-- undergraduates, reporters, politicians-- hear that the richest 5% of the people have half the total wealth, they tend to think *injustice!* an experienced programmer would be more likely to think *is that all?* the top 5% of programmers probably write 99% of the good software.

wealth can be created without being sold. scientists, till recently at least, effectively donated the wealth they created. we are all richer for knowing about penicillin, because we're less likely to die from infections. wealth is whatever people want, and not dying is certainly something we want. hackers often donate their work by writing open source software that anyone can use for free. i am much the richer for the operating system freebsd, which i'm running on the computer i'm using now, and so is yahoo, which runs it on all their servers.

**what a job is**

in industrialized countries, people belong to one institution or another at least until their twenties. after all those years you get used to the idea of belonging to a group of people who all get up in the morning, go to some set of buildings, and do things that they do not, ordinarily, enjoy doing. belonging to such a group becomes part of your identity: name, age, role, institution. if you have to introduce

yourself, or someone else describes you, it will be as something like, john smith, age 10, a student at such and such elementary school, or john smith, age 20, a student at such and such college.

when john smith finishes school he is expected to get a job. and what getting a job seems to mean is joining another institution. superficially it's a lot like college. you pick the companies you want to work for and apply to join them. if one likes you, you become a member of this new group. you get up in the morning and go to a new set of buildings, and do things that you do not, ordinarily, enjoy doing. there are a few differences: life is not as much fun, and you get paid, instead of paying, as you did in college. but the similarities feel greater than the differences. john smith is now john smith, 22, a software developer at such and such corporation.

in fact john smith's life has changed more than he realizes. socially, a company looks much like college, but the deeper you go into the underlying reality, the more different it gets.

what a company does, and has to do if it wants to continue to exist, is earn money. and the way most companies make money is by creating wealth. companies can be so specialized that this similarity is concealed, but it is not only manufacturing companies that create wealth. a big component of wealth is location. remember that magic machine that could make you cars and cook you dinner and so on? it would not be so useful if it delivered your dinner to a random location in central asia. if wealth means what people want, companies that move things also create wealth. ditto for many other kinds of companies that don't make anything physical. nearly all companies exist to do something people want.

and that's what you do, as well, when you go to work for a company. but here there is another layer that tends to obscure the underlying reality. in a company, the work you do is averaged together with a lot of other people's. you may not even be aware you're doing something people want. your contribution may be indirect. but the company as a whole must be giving people something they want, or they won't make any money. and if they are paying you x dollars a year, then on average you must be contributing at least x dollars a year worth of work, or the company will be spending more than it makes, and will go out of business.

someone graduating from college thinks, and is told, that he needs to get a job, as if the important thing were becoming a member of an institution. a more direct way to put it would be: you need to start doing something people want. you don't need to join a company to do that. all a company is is a group of people working together to do something people want. it's doing something people want that matters, not joining the group.[6]

for most people the best plan probably is to go to work for some existing company. but it is a good idea to understand what's happening when you do this. a job means doing something people want, averaged together with everyone else in that company.

**working harder**

that averaging gets to be a problem. i think the single biggest problem afflicting large companies is the difficulty of assigning a value to each person's work. for the most part they punt. in a big company you get paid a fairly predictable salary for working fairly hard. you're expected not to be obviously incompetent or lazy, but you're not expected to devote your whole life to your work.

it turns out, though, that there are economies of scale in how much of your life you devote to your work. in the right kind of business, someone who really devoted himself to work could generate ten or even a hundred times as much wealth as an average employee. a programmer, for example, instead of chugging along maintaining and updating an existing piece of software, could write a whole new piece of software, and with it create a new source of revenue.

companies are not set up to reward people who want to do this. you can't go to your boss and say, i'd like to start working ten times as hard, so will you please pay me ten times as much? for one thing, the official fiction is that you are already working as hard as you can. but a more serious problem is that the company has no way of measuring the value of your work.

salesmen are an exception. it's easy to measure how much revenue they generate, and they're usually paid a percentage of it. if a salesman wants to work harder, he can just start doing it, and he will automatically get paid proportionally more.

there is one other job besides sales where big companies can hire first-rate people: in the top management jobs. and for the same reason: their performance can be measured. the top managers are held responsible for the performance of the entire company. because an ordinary employee's performance can't usually be measured, he is not expected to do more than put in a solid effort. whereas top management, like salespeople, have to actually come up with the numbers. the ceo of a company that tanks cannot plead that he put in a solid effort. if the company does badly, he's done badly.

a company that could pay all its employees so straightforwardly would be enormously successful. many employees would work harder if they could get paid for it. more importantly, such a company would attract people who wanted to work especially hard. it would crush its competitors.

unfortunately, companies can't pay everyone like salesmen. salesmen work alone. most employees' work is tangled together. suppose a company makes some kind of consumer gadget. the engineers build a reliable gadget with all kinds of new features; the industrial designers design a beautiful case for it; and then the marketing people convince everyone that it's something they've got to have. how do you know how much of the gadget's sales are due to each group's efforts? or, for that matter, how much is due to the creators of past gadgets that gave the company a reputation for quality? there's no way to untangle all their contributions. even if you could read the minds of the consumers, you'd find these factors were all blurred together.

if you want to go faster, it's a problem to have your work tangled together with a large number of other people's. in a large group, your performance is not separately measurable-- and the rest of the group slows you down.

**measurement and leverage**

to get rich you need to get yourself in a situation with two things, measurement and leverage. you need to be in a position where your performance can be measured, or there is no way to get paid more by doing more. and you have to have leverage, in the sense that the decisions you make have a big effect.

measurement alone is not enough. an example of a job with measurement but not leverage is doing piecework in a sweatshop. your performance is measured and you get paid accordingly, but you have no scope for decisions. the only decision you get to make is how fast you work, and that can probably only increase your earnings by a factor of two or three.

an example of a job with both measurement and leverage would be lead actor in a movie. your performance can be measured in the gross of the movie. and you have leverage in the sense that your performance can make or break it.

ceos also have both measurement and leverage. they're measured, in that the performance of the company is their performance. and they have leverage in that their decisions set the whole company moving in one direction or another.

i think everyone who gets rich by their own efforts will be found to be in a situation with measurement and leverage. everyone i can think of does: ceos, movie stars, hedge fund managers, professional athletes. a good hint to the presence of leverage is the possibility of failure. upside must be balanced by downside, so if there is big potential for gain there must also be a terrifying possibility of loss. ceos, stars, fund managers, and athletes all live with the sword hanging over their heads; the moment they start to suck, they're out. if you're in a job that feels safe, you are not going to get rich, because if there is no danger there is almost certainly no leverage.

but you don't have to become a ceo or a movie star to be in a situation with measurement and leverage. all you need to do is be part of a small group working on a hard problem.

**smallness = measurement**

if you can't measure the value of the work done by individual employees, you can get close. you can measure the value of the work done by small groups.

one level at which you can accurately measure the revenue generated by employees is at the level of the whole company. when the company is small, you are thereby fairly close to measuring the contributions of individual employees. a viable startup might only have ten employees, which puts you within a factor of ten of measuring individual effort.

starting or joining a startup is thus as close as most people can get to saying to one's boss, i want to work ten times as hard, so please pay me ten times as much. there are two differences: you're not saying it to your boss, but directly to the customers (for whom your boss is only a proxy after all), and you're not doing it individually, but along with a small group of other ambitious people.

it will, ordinarily, be a group. except in a few unusual kinds of work, like acting or writing books, you can't be a company of one person. and the people you work with had better be good, because it's their work that yours is going to be averaged with.

a big company is like a giant galley driven by a thousand rowers. two things keep the speed of the galley down. one is that individual rowers don't see any result from working harder. the other is that, in a group of a thousand people, the average rower is likely to be pretty average.

if you took ten people at random out of the big galley and put them in a boat by themselves, they could probably go faster. they would have both carrot and stick to motivate them. an energetic rower would be encouraged by the thought that he could have a visible effect on the speed of the boat. and if someone was lazy, the others would be more likely to notice and complain.

but the real advantage of the ten-man boat shows when you take the ten *best* rowers out of the big galley and put them in a boat together. they will have all the extra motivation that comes from being in a small group. but more importantly, by selecting that small a group you can get the best rowers. each one will be in the top 1%. it's a much better deal for them to average their work together with a small group of their peers than to average it with everyone.

that's the real point of startups. ideally, you are getting together with a group of other people who also want to work a lot harder, and get paid a lot more, than they would in a big company. and because startups tend to get founded by self-selecting groups of ambitious people who already know one another (at least by reputation), the level of measurement is more precise than you get from smallness alone. a startup is not merely ten people, but ten people like you.

steve jobs once said that the success or failure of a startup depends on the first ten employees. i agree. if anything, it's more like the first five. being small is not, in itself, what makes startups kick butt, but

rather that small groups can be select. you don't want small in the sense of a village, but small in the sense of an all-star team.

the larger a group, the closer its average member will be to the average for the population as a whole. so all other things being equal, a very able person in a big company is probably getting a bad deal, because his performance is dragged down by the overall lower performance of the others. of course, all other things often are not equal: the able person may not care about money, or may prefer the stability of a large company. but a very able person who does care about money will ordinarily do better to go off and work with a small group of peers.

**technology = leverage**

startups offer anyone a way to be in a situation with measurement and leverage. they allow measurement because they're small, and they offer leverage because they make money by inventing new technology.

what is technology? it's *technique*. it's the way we all do things. and when you discover a new way to do things, its value is multiplied by all the people who use it. it is the proverbial fishing rod, rather than the fish. that's the difference between a startup and a restaurant or a barber shop. you fry eggs or cut hair one customer at a time. whereas if you solve a technical problem that a lot of people care about, you help everyone who uses your solution. that's leverage.

if you look at history, it seems that most people who got rich by creating wealth did it by developing new technology. you just can't fry eggs or cut hair fast enough. what made the florentines rich in 1200 was the discovery of new techniques for making the high-tech product of the time, fine woven cloth. what made the dutch rich in 1600 was the discovery of shipbuilding and navigation techniques that enabled them to dominate the seas of the far east.

fortunately there is a natural fit between smallness and solving hard problems. the leading edge of technology moves fast. technology that's valuable today could be worthless in a couple years. small companies are more at home in this world, because they don't have layers of bureaucracy to slow them down. also, technical advances tend to come from unorthodox approaches, and small companies are less constrained by convention.

big companies can develop technology. they just can't do it quickly. their size makes them slow and prevents them from rewarding employees for the extraordinary effort required. so in practice big companies only get to develop technology in fields where large capital requirements prevent startups from competing with them, like microprocessors, power plants, or passenger aircraft. and even in those fields they depend heavily on startups for components and ideas.

it's obvious that biotech or software startups exist to solve hard technical problems, but i think it will also be found to be true in businesses that don't seem to be about technology. mcdonald's, for example, grew big by designing a system, the mcdonald's franchise, that could then be reproduced at will all over the face of the earth. a mcdonald's franchise is controlled by rules so precise that it is practically a piece of software. write once, run everywhere. ditto for wal-mart. sam walton got rich not by being a retailer, but by designing a new kind of store.

use difficulty as a guide not just in selecting the overall aim of your company, but also at decision points along the way. at viaweb one of our rules of thumb was *run upstairs*. suppose you are a little, nimble guy being chased by a big, fat, bully. you open a door and find yourself in a staircase. do you go up or down? i say up. the bully can probably run downstairs as fast as you can. going upstairs his bulk will be more of a disadvantage. running upstairs is hard for you but even harder for him.

what this meant in practice was that we deliberately sought hard problems. if there were two features we could add to our software, both equally valuable in proportion to their difficulty, we'd always take the harder one. not just because it was more valuable, but *because it was harder.* we delighted in forcing bigger, slower competitors to follow us over difficult ground. like guerillas, startups prefer the difficult terrain of the mountains, where the troops of the central government can't follow. i can remember times when we were just exhausted after wrestling all day with some horrible technical problem. and i'd be delighted, because something that was hard for us would be impossible for our competitors.

this is not just a good way to run a startup. it's what a startup is. venture capitalists know about this and have a phrase for it: *barriers to entry.* if you go to a vc with a new idea and ask him to invest in it, one of the first things he'll ask is, how hard would this be for someone else to develop? that is, how much difficult ground have you put between yourself and potential pursuers?[7] and you had better have a convincing explanation of why your technology would be hard to duplicate. otherwise as soon as some big company becomes aware of it, they'll make their own, and with their brand name, capital, and distribution clout, they'll take away your market overnight. you'd be like guerillas caught in the open field by regular army forces.

one way to put up barriers to entry is through patents. but patents may not provide much protection. competitors commonly find ways to work around a patent. and if they can't, they may simply violate it and invite you to sue them. a big company is not afraid to be sued; it's an everyday thing for them. they'll make sure that suing them is expensive and takes a long time. ever heard of philo farnsworth? he invented television. the reason you've never heard of him is that his company was not the one to make money from it.[8] the company that did was rca, and farnsworth's reward for his efforts was a decade of patent litigation.

here, as so often, the best defense is a good offense. if you can develop technology that's simply too hard for competitors to duplicate, you don't need to rely on other defenses. start by picking a hard problem, and then at every decision point, take the harder choice.[9]

**the catch(es)**

if it were simply a matter of working harder than an ordinary employee and getting paid proportionately, it would obviously be a good deal to start a startup. up to a point it would be more fun. i don't think many people like the slow pace of big companies, the interminable meetings, the water-cooler conversations, the clueless middle managers, and so on.

unfortunately there are a couple catches. one is that you can't choose the point on the curve that you want to inhabit. you can't decide, for example, that you'd like to work just two or three times as hard, and get paid that much more. when you're running a startup, your competitors decide how hard you work. and they pretty much all make the same decision: as hard as you possibly can.

the other catch is that the payoff is only on average proportionate to your productivity. there is, as i said before, a large random multiplier in the success of any company. so in practice the deal is not that you're 30 times as productive and get paid 30 times as much. it is that you're 30 times as productive, and get paid between zero and a thousand times as much. if the mean is 30x, the median is probably zero. most startups tank, and not just the dogfood portals we all heard about during the internet bubble. it's common for a startup to be developing a genuinely good product, take slightly too long to do it, run out of money, and have to shut down.

a startup is like a mosquito. a bear can absorb a hit and a crab is armored against one, but a mosquito is designed for one thing: to score. no energy is wasted on defense. the defense of mosquitos, as a

species, is that there are a lot of them, but this is little consolation to the individual mosquito.

startups, like mosquitos, tend to be an all-or-nothing proposition. and you don't generally know which of the two you're going to get till the last minute. viaweb came close to tanking several times. our trajectory was like a sine wave. fortunately we got bought at the top of the cycle, but it was damned close. while we were visiting yahoo in california to talk about selling the company to them, we had to borrow a conference room to reassure an investor who was about to back out of a new round of funding that we needed to stay alive.

the all-or-nothing aspect of startups was not something we wanted. viaweb's hackers were all extremely risk-averse. if there had been some way just to work super hard and get paid for it, without having a lottery mixed in, we would have been delighted. we would have much preferred a 100% chance of $1 million to a 20% chance of $10 million, even though theoretically the second is worth twice as much. unfortunately, there is not currently any space in the business world where you can get the first deal.

the closest you can get is by selling your startup in the early stages, giving up upside (and risk) for a smaller but guaranteed payoff. we had a chance to do this, and stupidly, as we then thought, let it slip by. after that we became comically eager to sell. for the next year or so, if anyone expressed the slightest curiosity about viaweb we would try to sell them the company. but there were no takers, so we had to keep going.

it would have been a bargain to buy us at an early stage, but companies doing acquisitions are not looking for bargains. a company big enough to acquire startups will be big enough to be fairly conservative, and within the company the people in charge of acquisitions will be among the more conservative, because they are likely to be business school types who joined the company late. they would rather overpay for a safe choice. so it is easier to sell an established startup, even at a large premium, than an early-stage one.

**get users**

i think it's a good idea to get bought, if you can. running a business is different from growing one. it is just as well to let a big company take over once you reach cruising altitude. it's also financially wiser, because selling allows you to diversify. what would you think of a financial advisor who put all his client's assets into one volatile stock?

how do you get bought? mostly by doing the same things you'd do if you didn't intend to sell the company. being profitable, for example. but getting bought is also an art in its own right, and one that we spent a lot of time trying to master.

potential buyers will always delay if they can. the hard part about getting bought is getting them to act. for most people, the most powerful motivator is not the hope of gain, but the fear of loss. for potential acquirers, the most powerful motivator is the prospect that one of their competitors will buy you. this, as we found, causes ceos to take red-eyes. the second biggest is the worry that, if they don't buy you now, you'll continue to grow rapidly and will cost more to acquire later, or even become a competitor.

in both cases, what it all comes down to is users. you'd think that a company about to buy you would do a lot of research and decide for themselves how valuable your technology was. not at all. what they go by is the number of users you have.

in effect, acquirers assume the customers know who has the best technology. and this is not as stupid as it sounds. users are the only real proof that you've created wealth. wealth is what people want, and

if people aren't using your software, maybe it's not just because you're bad at marketing. maybe it's because you haven't made what they want.

venture capitalists have a list of danger signs to watch out for. near the top is the company run by techno-weenies who are obsessed with solving interesting technical problems, instead of making users happy. in a startup, you're not just trying to solve problems. you're trying to solve problems *that users care about*.

so i think you should make users the test, just as acquirers do. treat a startup as an optimization problem in which performance is measured by number of users. as anyone who has tried to optimize software knows, the key is measurement. when you try to guess where your program is slow, and what would make it faster, you almost always guess wrong.

number of users may not be the perfect test, but it will be very close. it's what acquirers care about. it's what revenues depend on. it's what makes competitors unhappy. it's what impresses reporters, and potential new users. certainly it's a better test than your a priori notions of what problems are important to solve, no matter how technically adept you are.

among other things, treating a startup as an optimization problem will help you avoid another pitfall that vcs worry about, and rightly-- taking a long time to develop a product. now we can recognize this as something hackers already know to avoid: premature optimization. get a version 1.0 out there as soon as you can. until you have some users to measure, you're optimizing based on guesses.

the ball you need to keep your eye on here is the underlying principle that wealth is what people want. if you plan to get rich by creating wealth, you have to know what people want. so few businesses really pay attention to making customers happy. how often do you walk into a store, or call a company on the phone, with a feeling of dread in the back of your mind? when you hear "your call is important to us, please stay on the line," do you think, oh good, now everything will be all right?

a restaurant can afford to serve the occasional burnt dinner. but in technology, you cook one thing and that's what everyone eats. so any difference between what people want and what you deliver is multiplied. you please or annoy customers wholesale. the closer you can get to what they want, the more wealth you generate.

**wealth and power**

making wealth is not the only way to get rich. for most of human history it has not even been the most common. until a few centuries ago, the main sources of wealth were mines, slaves and serfs, land, and cattle, and the only ways to acquire these rapidly were by inheritance, marriage, conquest, or confiscation. naturally wealth had a bad reputation.

two things changed. the first was the rule of law. for most of the world's history, if you did somehow accumulate a fortune, the ruler or his henchmen would find a way to steal it. but in medieval europe something new happened. a new class of merchants and manufacturers began to collect in towns.[10] together they were able to withstand the local feudal lord. so for the first time in our history, the bullies stopped stealing the nerds' lunch money. this was naturally a great incentive, and possibly indeed the main cause of the second big change, industrialization.

a great deal has been written about the causes of the industrial revolution. but surely a necessary, if not sufficient, condition was that people who made fortunes be able to enjoy them in peace.[11] one piece of evidence is what happened to countries that tried to return to the old model, like the soviet union, and

to a lesser extent britain under the labor governments of the 1960s and early 1970s. take away the incentive of wealth, and technical innovation grinds to a halt.

remember what a startup is, economically: a way of saying, i want to work faster. instead of accumulating money slowly by being paid a regular wage for fifty years, i want to get it over with as soon as possible. so governments that forbid you to accumulate wealth are in effect decreeing that you work slowly. they're willing to let you earn $3 million over fifty years, but they're not willing to let you work so hard that you can do it in two. they are like the corporate boss that you can't go to and say, i want to work ten times as hard, so please pay me ten times a much. except this is not a boss you can escape by starting your own company.

the problem with working slowly is not just that technical innovation happens slowly. it's that it tends not to happen at all. it's only when you're deliberately looking for hard problems, as a way to use speed to the greatest advantage, that you take on this kind of project. developing new technology is a pain in the ass. it is, as edison said, one percent inspiration and ninety-nine percent perspiration. without the incentive of wealth, no one wants to do it. engineers will work on sexy projects like fighter planes and moon rockets for ordinary salaries, but more mundane technologies like light bulbs or semiconductors have to be developed by entrepreneurs.

startups are not just something that happened in silicon valley in the last couple decades. since it became possible to get rich by creating wealth, everyone who has done it has used essentially the same recipe: measurement and leverage, where measurement comes from working with a small group, and leverage from developing new techniques. the recipe was the same in florence in 1200 as it is in santa clara today.

understanding this may help to answer an important question: why europe grew so powerful. was it something about the geography of europe? was it that europeans are somehow racially superior? was it their religion? the answer (or at least the proximate cause) may be that the europeans rode on the crest of a powerful new idea: allowing those who made a lot of money to keep it.

once you're allowed to do that, people who want to get rich can do it by generating wealth instead of stealing it. the resulting technological growth translates not only into wealth but into military power. the theory that led to the stealth plane was developed by a soviet mathematician. but because the soviet union didn't have a computer industry, it remained for them a theory; they didn't have hardware capable of executing the calculations fast enough to design an actual airplane.

in that respect the cold war teaches the same lesson as world war ii and, for that matter, most wars in recent history. don't let a ruling class of warriors and politicians squash the entrepreneurs. the same recipe that makes individuals rich makes countries powerful. let the nerds keep their lunch money, and you rule the world.

---

## notes

1 one valuable thing you tend to get only in startups is *uninterruptability*. different kinds of work have different time quanta. someone proofreading a manuscript could probably be interrupted every fifteen minutes with little loss of productivity. but the time quantum for hacking is very long: it might take an hour just to load a problem into your head. so the cost of having someone from personnel call you about a form you forgot to fill out can be huge.

this is why hackers give you such a baleful stare as they turn from their screen to answer your question. inside their heads a giant house of cards is tottering.

the mere possibility of being interrupted deters hackers from starting hard projects. this is why they tend to work late at night, and why it's next to impossible to write great software in a cubicle (except late at night).

one great advantage of startups is that they don't yet have any of the people who interrupt you. there is no personnel department, and thus no form nor anyone to call you about it.

2 faced with the idea that people working for startups might be 20 or 30 times as productive as those working for large companies, executives at large companies will naturally wonder, how could i get the people working for me to do that? the answer is simple: pay them to.

internally most companies are run like communist states. if you believe in free markets, why not turn your company into one?

hypothesis: a company will be maximally profitable when each employee is paid in proportion to the wealth they generate.

3 until recently even governments sometimes didn't grasp the distinction between money and wealth. adam smith (*wealth of nations*, v:i) mentions several that tried to preserve their "wealth" by forbidding the export of gold or silver. but having more of the medium of exchange would not make a country richer; if you have more money chasing the same amount of material wealth, the only result is higher prices.

4 there are many senses of the word "wealth," not all of them material. i'm not trying to make a deep philosophical point here about which is the true kind. i'm writing about one specific, rather technical sense of the word "wealth." what people will give you money for. this is an interesting sort of wealth to study, because it is the kind that prevents you from starving. and what people will give you money for depends on them, not you.

when you're starting a business, it's easy to slide into thinking that customers want what you do. during the internet bubble i talked to a woman who, because she liked the outdoors, was starting an "outdoor portal." you know what kind of business you should start if you like the outdoors? one to recover data from crashed hard disks.

what's the connection? none at all. which is precisely my point. if you want to create wealth (in the narrow technical sense of not starving) then you should be especially skeptical about any plan that centers on things you like doing. that is where your idea of what's valuable is least likely to coincide with other people's.

5 in the average car restoration you probably do make everyone else microscopically poorer, by doing a small amount of damage to the environment. while environmental costs should be taken into account, they don't make wealth a zero-sum game. for example, if you repair a machine that's broken because a part has come unscrewed, you create wealth with no environmental cost.

[5b] this essay was written before firefox.

6 many people feel confused and depressed in their early twenties. life seemed so much more fun in college. well, of course it was. don't be fooled by the surface similarities. you've gone from guest to servant. it's possible to have fun in this new world. among other things, you now get to go behind the doors that say "authorized personnel only." but the change is a shock at first, and all the worse if you're not consciously aware of it.

7 when vcs asked us how long it would take another startup to duplicate our software, we used to reply that they probably wouldn't be able to at all. i think this made us seem naive, or liars.

8 few technologies have one clear inventor. so as a rule, if you know the "inventor" of something (the telephone, the assembly line, the airplane, the light bulb, the transistor) it is because their company made money from it, and the company's pr people worked hard to spread the story. if you don't know who invented somet-

hing (the automobile, the television, the computer, the jet engine, the laser), it's because other companies made all the money.

9 this is a good plan for life in general. if you have two choices, choose the harder. if you're trying to decide whether to go out running or sit home and watch tv, go running. probably the reason this trick works so well is that when you have two choices and one is harder, the only reason you're even considering the other is laziness. you know in the back of your mind what's the right thing to do, and this trick merely forces you to acknowledge it.

10 it is probably no accident that the middle class first appeared in northern italy and the low countries, where there were no strong central governments. these two regions were the richest of their time and became the twin centers from which renaissance civilization radiated. if they no longer play that role, it is because other places, like the united states, have been truer to the principles they discovered.

11 it may indeed be a sufficient condition. but if so, why didn't the industrial revolution happen earlier? two possible (and not incompatible) answers: (a) it did. the industrial revolution was one in a series. (b) because in medieval towns, monopolies and guild regulations initially slowed the development of new means of production.

comment on this essay.

02

# how people get rich now

april 2021
paulgraham.com/richnow.html

every year since 1982, *forbes* magazine has published a list of the richest americans. if we compare the 100 richest people in 1982 to the 100 richest in 2020, we notice some big differences.

in 1982 the most common source of wealth was inheritance. of the 100 richest people, 60 inherited from an ancestor. there were 10 du pont heirs alone. by 2020 the number of heirs had been cut in half, accounting for only 27 of the biggest 100 fortunes.

why would the percentage of heirs decrease? not because inheritance taxes increased. in fact, they decreased significantly during this period. the reason the percentage of heirs has decreased is not that fewer people are inheriting great fortunes, but that more people are making them.

how are people making these new fortunes? roughly 3/4 by starting companies and 1/4 by investing. of the 73 new fortunes in 2020, 56 derive from founders' or early employees' equity (52 founders, 2 early employees, and 2 wives of founders), and 17 from managing investment funds.

there were no fund managers among the 100 richest americans in 1982. hedge funds and private equity firms existed in 1982, but none of their founders were rich enough yet to make it into the top 100. two things changed: fund managers discovered new ways to generate high returns, and more investors were willing to trust them with their money. [1]

but the main source of new fortunes now is starting companies, and when you look at the data, you see big changes there too. people get richer from starting companies now than they did in 1982, because the companies do different things.

in 1982, there were two dominant sources of new wealth: oil and real estate. of the 40 new fortunes in 1982, at least 24 were due primarily to oil or real estate. now only a small number are: of the 73 new fortunes in 2020, 4 were due to real estate and only 2 to oil.

by 2020 the biggest source of new wealth was what are sometimes called "tech" companies. of the 73 new fortunes, about 30 derive from such companies. these are particularly common among the richest of the rich: 8 of the top 10 fortunes in 2020 were new fortunes of this type.

arguably it's slightly misleading to treat tech as a category. isn't amazon really a retailer, and tesla a car maker? yes and no. maybe in 50 years, when what we call tech is taken for granted, it won't seem right to put these two businesses in the same category. but at the moment at least, there is definitely something they share in common that distinguishes them. what retailer starts aws? what car maker is run by someone who also has a rocket company?

the tech companies behind the top 100 fortunes also form a well-differentiated group in the sense that they're all companies that venture capitalists would readily invest in, and the others mostly not. and there's a reason why: these are mostly companies that win by having better technology, rather than just a ceo who's really driven and good at making deals.

to that extent, the rise of the tech companies represents a qualitative change. the oil and real estate magnates of the 1982 forbes 400 didn't win by making better technology. they won by being really driven and good at making deals. [2] and indeed, that way of getting rich is so old that it predates the industrial revolution. the courtiers who got rich in the (nominal) service of european royal houses in the 16th and 17th centuries were also, as a rule, really driven and good at making deals.

people who don't look any deeper than the gini coefficient look back on the world of 1982 as the good old days, because those who got rich then didn't get as rich. but if you dig into *how* they got rich, the old days don't look so good. in 1982, 84% of the richest 100 people got rich by inheritance, extracting

natural resources, or doing real estate deals. is that really better than a world in which the richest people get rich by starting tech companies?

why are people starting so many more new companies than they used to, and why are they getting so rich from it? the answer to the first question, curiously enough, is that it's misphrased. we shouldn't be asking why people are starting companies, but why they're starting companies *again*. [3]

in 1892, the *new york herald tribune* compiled a list of all the millionaires in america. they found 4047 of them. how many had inherited their wealth then? only about 20%, which is less than the proportion of heirs today. and when you investigate the sources of the new fortunes, 1892 looks even more like today. hugh rockoff found that "many of the richest ... gained their initial edge from the new technology of mass production." [4]

so it's not 2020 that's the anomaly here, but 1982. the real question is why so few people had gotten rich from starting companies in 1982. and the answer is that even as the *herald tribune*'s list was being compiled, a wave of consolidation was sweeping through the american economy. in the late 19th and early 20th centuries, financiers like j. p. morgan combined thousands of smaller companies into a few hundred giant ones with commanding economies of scale. by the end of world war ii, as michael lind writes, "the major sectors of the economy were either organized as government-backed cartels or dominated by a few oligopolistic corporations." [5]

in 1960, most of the people who start startups today would have gone to work for one of them. you could get rich from starting your own company in 1890 and in 2020, but in 1960 it was not really a viable option. you couldn't break through the oligopolies to get at the markets. so the prestigious route in 1960 was not to start your own company, but to work your way up the corporate ladder at an existing one. [6]

making everyone a corporate employee decreased economic inequality (and every other kind of variation), but if your model of normal is the mid 20th century, you have a very misleading model in that respect. j. p. morgan's economy turned out to be just a phase, and starting in the 1970s, it began to break up.

why did it break up? partly senescence. the big companies that seemed models of scale and efficiency in 1930 had by 1970 become slack and bloated. by 1970 the rigid structure of the economy was full of cosy nests that various groups had built to insulate themselves from market forces. during the carter administration the federal government realized something was amiss and began, in a process they called "deregulation," to roll back the policies that propped up the oligopolies.

but it wasn't just decay from within that broke up j. p. morgan's economy. there was also pressure from without, in the form of new technology, and particularly microelectronics. the best way to envision what happened is to imagine a pond with a crust of ice on top. initially the only way from the bottom to the surface is around the edges. but as the ice crust weakens, you start to be able to punch right through the middle.

the edges of the pond were pure tech: companies that actually described themselves as being in the electronics or software business. when you used the word "startup" in 1990, that was what you meant. but now startups are punching right through the middle of the ice crust and displacing incumbents like retailers and tv networks and car companies. [7]

but though the breakup of j. p. morgan's economy created a new world in the technological sense, it was a reversion to the norm in the social sense. if you only look back as far as the mid 20th century, it seems like people getting rich by starting their own companies is a recent phenomenon. but if you look

back further, you realize it's actually the default. so what we should expect in the future is more of the same. indeed, we should expect both the number and wealth of founders to grow, because every decade it gets easier to start a startup.

part of the reason it's getting easier to start a startup is social. society is (re)assimilating the concept. if you start one now, your parents won't freak out the way they would have a generation ago, and knowledge about how to do it is much more widespread. but the main reason it's easier to start a startup now is that it's cheaper. technology has driven down the cost of both building products and acquiring customers.

the decreasing cost of starting a startup has in turn changed the balance of power between founders and investors. back when starting a startup meant building a factory, you needed investors' permission to do it at all. but now investors need founders more than founders need investors, and that, combined with the increasing amount of venture capital available, has driven up valuations. [8]

so the decreasing cost of starting a startup increases the number of rich people in two ways: it means that more people start them, and that those who do can raise money on better terms.

but there's also a third factor at work: the companies themselves are more valuable, because newly founded companies grow faster than they used to. technology hasn't just made it cheaper to build and distribute things, but faster too.

this trend has been running for a long time. ibm, founded in 1896, took 45 years to reach a billion 2020 dollars in revenue. hewlett-packard, founded in 1939, took 25 years. microsoft, founded in 1975, took 13 years. now the norm for fast-growing companies is 7 or 8 years. [9]

fast growth has a double effect on the value of founders' stock. the value of a company is a function of its revenue and its growth rate. so if a company grows faster, you not only get to a billion dollars in revenue sooner, but the company is more valuable when it reaches that point than it would be if it were growing slower.

that's why founders sometimes get so rich so young now. the low initial cost of starting a startup means founders can start young, and the fast growth of companies today means that if they succeed they could be surprisingly rich just a few years later.

it's easier now to start and grow a company than it has ever been. that means more people start them, that those who do get better terms from investors, and that the resulting companies become more valuable. once you understand how these mechanisms work, and that startups were suppressed for most of the 20th century, you don't have to resort to some vague right turn the country took under reagan to explain why america's gini coefficient is increasing. of course the gini coefficient is increasing. with more people starting more valuable companies, how could it not be?

---

## notes

1 investment firms grew rapidly after a regulatory change by the labor department in 1978 allowed pension funds to invest in them, but the effects of this growth were not yet visible in the top 100 fortunes in 1982.

2 george mitchell deserves mention as an exception. though really driven and good at making deals, he was also the first to figure out how to use fracking to get natural gas out of shale.

3 when i say people are starting more companies, i mean the type of company meant to grow very big. there has actually been a decrease in the last couple decades in the overall number of new companies. but the vast majority of companies are small retail and service businesses. so what the statistics about the decreasing number of new businesses mean is that people are starting fewer shoe stores and barber shops.

people sometimes get confused when they see a graph labelled "startups" that's going down, because there are two senses of the word "startup": (1) the founding of a company, and (2) a particular type of company designed to grow big fast. the statistics mean startup in sense (1), not sense (2).

4 rockoff, hugh. "great fortunes of the gilded age." nber working paper 14555, 2008.

5 lind, michael. *land of promise.* harpercollins, 2012.

it's also likely that the high tax rates in the mid 20th century deterred people from starting their own companies. starting one's own company is risky, and when risk isn't rewarded, people opt for safety instead.

but it wasn't simply cause and effect. the oligopolies and high tax rates of the mid 20th century were all of a piece. lower taxes are not just a cause of entrepreneurship, but an effect as well: the people getting rich in the mid 20th century from real estate and oil exploration lobbied for and got huge tax loopholes that made their effective tax rate much lower, and presumably if it had been more common to grow big companies by building new technology, the people doing that would have lobbied for their own loopholes as well.

6 that's why the people who did get rich in the mid 20th century so often got rich from oil exploration or real estate. those were the two big areas of the economy that weren't susceptible to consolidation.

7 the pure tech companies used to be called "high technology" startups. but now that startups can punch through the middle of the ice crust, we don't need a separate name for the edges, and the term "high-tech" has a decidedly retro sound.

8 higher valuations mean you either sell less stock to get a given amount of money, or get more money for a given amount of stock. the typical startup does some of each. obviously you end up richer if you keep more stock, but you should also end up richer if you raise more money, because (a) it should make the company more successful, and (b) you should be able to last longer before the next round, or not even need one. notice all those shoulds though. in practice a lot of money slips through them.

it might seem that the huge rounds raised by startups nowadays contradict the claim that it has become cheaper to start one. but there's no contradiction here; the startups that raise the most are the ones doing it by choice, in order to grow faster, not the ones doing it because they need the money to survive. there's nothing like not needing money to make people offer it to you.

you would think, after having been on the side of labor in its fight with capital for almost two centuries, that the far left would be happy that labor has finally prevailed. but none of them seem to be. you can almost hear them saying "no, no, not *that* way."

9 ibm was created in 1911 by merging three companies, the most important of which was herman hollerith's tabulating machine company, founded in 1896. in 1941 its revenues were $60 million.

hewlett-packard's revenues in 1964 were $125 million.

microsoft's revenues in 1988 were $590 million.

**thanks** to trevor blackwell, jessica livingston, bob lesko, robert morris, russ roberts, and alex tabarrok for reading drafts of this, and to jon erlichman for growth data.

03

# billionaires build

december 2020
paulgraham.com/ace.html

as i was deciding what to write about next, i was surprised to find that two separate essays i'd been planning to write were actually the same.

the first is about how to ace your y combinator interview. there has been so much nonsense written about this topic that i've been meaning for years to write something telling founders the truth.

the second is about something politicians sometimes say — that the only way to become a billionaire is by exploiting people — and why this is mistaken.

keep reading, and you'll learn both simultaneously.

i know the politicians are mistaken because it was my job to predict which people will become billionaires. i think i can truthfully say that i know as much about how to do this as anyone. if the key to becoming a billionaire — the defining feature of billionaires — was to exploit people, then i, as a professional billionaire scout, would surely realize this and look for people who would be good at it, just as an nfl scout looks for speed in wide receivers.

but aptitude for exploiting people is not what y combinator looks for at all. in fact, it's the opposite of what they look for. i'll tell you what they do look for, by explaining how to convince y combinator to fund you, and you can see for yourself.

what yc looks for, above all, is founders who understand some group of users and can make what they want. this is so important that it's yc's motto: "make something people want."

a big company can to some extent force unsuitable products on unwilling customers, but a startup doesn't have the power to do that. a startup must sing for its supper, by making things that genuinely delight its customers. otherwise it will never get off the ground.

here's where things get difficult, both for you as a founder and for the yc partners trying to decide whether to fund you. in a market economy, it's hard to make something people want that they don't already have. that's the great thing about market economies. if other people both knew about this need and were able to satisfy it, they already would be, and there would be no room for your startup.

which means the conversation during your yc interview will have to be about something new: either a new need, or a new way to satisfy one. and not just new, but uncertain. if it were certain that the need existed and that you could satisfy it, that certainty would be reflected in large and rapidly growing revenues, and you wouldn't be seeking seed funding.

so the yc partners have to guess both whether you've discovered a real need, and whether you'll be able to satisfy it. that's what they are, at least in this part of their job: professional guessers. they have 1001 heuristics for doing this, and i'm not going to tell you all of them, but i'm happy to tell you the most important ones, because these can't be faked; the only way to "hack" them would be to do what you should be doing anyway as a founder.

the first thing the partners will try to figure out, usually, is whether what you're making will ever be something a lot of people want. it doesn't have to be something a lot of people want now. the product and the market will both evolve, and will influence each other's evolution. but in the end there has to be something with a huge market. that's what the partners will be trying to figure out: is there a path to a huge market? [1]

sometimes it's obvious there will be a huge market. if boom manages to ship an airliner at all, international airlines will have to buy it. but usually it's not obvious. usually the path to a huge market is

by growing a small market. this idea is important enough that it's worth coining a phrase for, so let's call one of these small but growable markets a "larval market."

the perfect example of a larval market might be apple's market when they were founded in 1976. in 1976, not many people wanted their own computer. but more and more started to want one, till now every 10 year old on the planet wants a computer (but calls it a "phone").

the ideal combination is the group of founders who are "living in the future" in the sense of being at the leading edge of some kind of change, and who are building something they themselves want. most super-successful startups are of this type. steve wozniak wanted a computer. mark zuckerberg wanted to engage online with his college friends. larry and sergey wanted to find things on the web. all these founders were building things they and their peers wanted, and the fact that they were at the leading edge of change meant that more people would want these things in the future.

but although the ideal larval market is oneself and one's peers, that's not the only kind. a larval market might also be regional, for example. you build something to serve one location, and then expand to others.

the crucial feature of the initial market is that it exist. that may seem like an obvious point, but the lack of it is the biggest flaw in most startup ideas. there have to be some people who want what you're building right now, and want it so urgently that they're willing to use it, bugs and all, even though you're a small company they've never heard of. there don't have to be many, but there have to be some. as long as you have some users, there are straightforward ways to get more: build new features they want, seek out more people like them, get them to refer you to their friends, and so on. but these techniques all require some initial seed group of users.

so this is one thing the yc partners will almost certainly dig into during your interview. who are your first users going to be, and how do you know they want this? if i had to decide whether to fund startups based on a single question, it would be "how do you know people want this?"

the most convincing answer is "because we and our friends want it." it's even better when this is followed by the news that you've already built a prototype, and even though it's very crude, your friends are using it, and it's spreading by word of mouth. if you can say that and you're not lying, the partners will switch from default no to default yes. meaning you're in unless there's some other disqualifying flaw.

that is a hard standard to meet, though. airbnb didn't meet it. they had the first part. they had made something they themselves wanted. but it wasn't spreading. so don't feel bad if you don't hit this gold standard of convincingness. if airbnb didn't hit it, it must be too high.

in practice, the yc partners will be satisfied if they feel that you have a deep understanding of your users' needs. and the airbnbs did have that. they were able to tell us all about what motivated hosts and guests. they knew from first-hand experience, because they'd been the first hosts. we couldn't ask them a question they didn't know the answer to. we ourselves were not very excited about the idea as users, but we knew this didn't prove anything, because there were lots of successful startups we hadn't been excited about as users. we were able to say to ourselves "they seem to know what they're talking about. maybe they're onto something. it's not growing yet, but maybe they can figure out how to make it grow during yc." which they did, about three weeks into the batch.

the best thing you can do in a yc interview is to teach the partners about your users. so if you want to prepare for your interview, one of the best ways to do it is to go talk to your users and find out exactly what they're thinking. which is what you should be doing anyway.

this may sound strangely credulous, but the yc partners want to rely on the founders to tell them about the market. think about how vcs typically judge the potential market for an idea. they're not ordinarily domain experts themselves, so they forward the idea to someone who is, and ask for their opinion. yc doesn't have time to do this, but if the yc partners can convince themselves that the founders both (a) know what they're talking about and (b) aren't lying, they don't need outside domain experts. they can use the founders themselves as domain experts when evaluating their own idea.

this is why yc interviews aren't pitches. to give as many founders as possible a chance to get funded, we made interviews as short as we could: 10 minutes. that is not enough time for the partners to figure out, through the indirect evidence in a pitch, whether you know what you're talking about and aren't lying. they need to dig in and ask you questions. there's not enough time for sequential access. they need random access. [2]

the worst advice i ever heard about how to succeed in a yc interview is that you should take control of the interview and make sure to deliver the message you want to. in other words, turn the interview into a pitch. ⟨elaborate expletive⟩. it is so annoying when people try to do that. you ask them a question, and instead of answering it, they deliver some obviously prefabricated blob of pitch. it eats up 10 minutes really fast.

there is no one who can give you accurate advice about what to do in a yc interview except a current or former yc partner. people who've merely been interviewed, even successfully, have no idea of this, but interviews take all sorts of different forms depending on what the partners want to know about most. sometimes they're all about the founders, other times they're all about the idea. sometimes some very narrow aspect of the idea. founders sometimes walk away from interviews complaining that they didn't get to explain their idea completely. true, but they explained enough.

since a yc interview consists of questions, the way to do it well is to answer them well. part of that is answering them candidly. the partners don't expect you to know everything. but if you don't know the answer to a question, don't try to bullshit your way out of it. the partners, like most experienced investors, are professional bullshit detectors, and you are (hopefully) an amateur bullshitter. and if you try to bullshit them and fail, they may not even tell you that you failed. so it's better to be honest than to try to sell them. if you don't know the answer to a question, say you don't, and tell them how you'd go about finding it, or tell them the answer to some related question.

if you're asked, for example, what could go wrong, the worst possible answer is "nothing." instead of convincing them that your idea is bullet-proof, this will convince them that you're a fool or a liar. far better to go into gruesome detail. that's what experts do when you ask what could go wrong. the partners know that your idea is risky. that's what a good bet looks like at this stage: a tiny probability of a huge outcome.

ditto if they ask about competitors. competitors are rarely what kills startups. poor execution does. but you should know who your competitors are, and tell the yc partners candidly what your relative strengths and weaknesses are. because the yc partners know that competitors don't kill startups, they won't hold competitors against you too much. they will, however, hold it against you if you seem either to be unaware of competitors, or to be minimizing the threat they pose. they may not be sure whether you're clueless or lying, but they don't need to be.

the partners don't expect your idea to be perfect. this is seed investing. at this stage, all they can expect are promising hypotheses. but they do expect you to be thoughtful and honest. so if trying to make

your idea seem perfect causes you to come off as glib or clueless, you've sacrificed something you needed for something you didn't.

if the partners are sufficiently convinced that there's a path to a big market, the next question is whether you'll be able to find it. that in turn depends on three things: the general qualities of the founders, their specific expertise in this domain, and the relationship between them. how determined are the founders? are they good at building things? are they resilient enough to keep going when things go wrong? how strong is their friendship?

though the airbnbs only did ok in the idea department, they did spectacularly well in this department. the story of how they'd funded themselves by making obama- and mccain-themed breakfast cereal was the single most important factor in our decision to fund them. they didn't realize it at the time, but what seemed to them an irrelevant story was in fact fabulously good evidence of their qualities as founders. it showed they were resourceful and determined, and could work together.

it wasn't just the cereal story that showed that, though. the whole interview showed that they cared. they weren't doing this just for the money, or because startups were cool. the reason they were working so hard on this company was because it was their project. they had discovered an interesting new idea, and they just couldn't let it go.

mundane as it sounds, that's the most powerful motivator of all, not just in startups, but in most ambitious undertakings: to be genuinely interested in what you're building. this is what really drives billionaires, or at least the ones who become billionaires from starting companies. the company is their project.

one thing few people realize about billionaires is that all of them could have stopped sooner. they could have gotten acquired, or found someone else to run the company. many founders do. the ones who become really rich are the ones who keep working. and what makes them keep working is not just money. what keeps them working is the same thing that keeps anyone else working when they could stop if they wanted to: that there's nothing else they'd rather do.

that, not exploiting people, is the defining quality of people who become billionaires from starting companies. so that's what yc looks for in founders: authenticity. people's motives for starting startups are usually mixed. they're usually doing it from some combination of the desire to make money, the desire to seem cool, genuine interest in the problem, and unwillingness to work for someone else. the last two are more powerful motivators than the first two. it's ok for founders to want to make money or to seem cool. most do. but if the founders seem like they're doing it *just* to make money or *just* to seem cool, they're not likely to succeed on a big scale. the founders who are doing it for the money will take the first sufficiently large acquisition offer, and the ones who are doing it to seem cool will rapidly discover that there are much less painful ways of seeming cool. [3]

y combinator certainly sees founders whose m.o. is to exploit people. yc is a magnet for them, because they want the yc brand. but when the yc partners detect someone like that, they reject them. if bad people made good founders, the yc partners would face a moral dilemma. fortunately they don't, because bad people make bad founders. this exploitative type of founder is not going to succeed on a large scale, and in fact probably won't even succeed on a small one, because they're always going to be taking shortcuts. they see yc itself as a shortcut.

their exploitation usually begins with their own cofounders, which is disastrous, since the cofounders' relationship is the foundation of the company. then it moves on to the users, which is also disastrous, because the sort of early adopters a successful startup wants as its initial users are the hardest to fool.

the best this kind of founder can hope for is to keep the edifice of deception tottering along until some acquirer can be tricked into buying it. but that kind of acquisition is never very big. [4]

if professional billionaire scouts know that exploiting people is not the skill to look for, why do some politicians think this is the defining quality of billionaires?

i think they start from the feeling that it's wrong that one person could have so much more money than another. it's understandable where that feeling comes from. it's in our dna, and even in the dna of other species.

if they limited themselves to saying that it made them feel bad when one person had so much more money than other people, who would disagree? it makes me feel bad too, and i think people who make a lot of money have a moral obligation to use it for the common good. the mistake they make is to jump from feeling bad that some people are much richer than others to the conclusion that there's no legitimate way to make a very large amount of money. now we're getting into statements that are not only falsifiable, but false.

there are certainly some people who become rich by doing bad things. but there are also plenty of people who behave badly and don't make that much from it. there is no correlation — in fact, probably an inverse correlation — between how badly you behave and how much money you make.

the greatest danger of this nonsense may not even be that it sends policy astray, but that it misleads ambitious people. can you imagine a better way to destroy social mobility than by telling poor kids that the way to get rich is by exploiting people, while the rich kids know, from having watched the preceding generation do it, how it's really done?

i'll tell you how it's really done, so you can at least tell your own kids the truth. it's all about users. the most reliable way to become a billionaire is to start a company that grows fast, and the way to grow fast is to make what users want. newly started startups have no choice but to delight users, or they'll never even get rolling. but this never stops being the lodestar, and bigger companies take their eye off it at their peril. stop delighting users, and eventually someone else will.

users are what the partners want to know about in yc interviews, and what i want to know about when i talk to founders that we funded ten years ago and who are billionaires now. what do users want? what new things could you build for them? founders who've become billionaires are always eager to talk about that topic. that's how they became billionaires.

---

## notes

1 the yc partners have so much practice doing this that they sometimes see paths that the founders themselves haven't seen yet. the partners don't try to seem skeptical, as buyers in transactions often do to increase their leverage. although the founders feel their job is to convince the partners of the potential of their idea, these roles are not infrequently reversed, and the founders leave the interview feeling their idea has more potential than they realized.

2 in practice, 7 minutes would be enough. you rarely change your mind at minute 8. but 10 minutes is socially convenient.

3 i myself took the first sufficiently large acquisition offer in my first startup, so i don't blame founders for doing this. there's nothing wrong with starting a startup to make money. you need to make money somehow, and

for some people startups are the most efficient way to do it. i'm just saying that these are not the startups that get really big.

4 not these days, anyway. there were some big ones during the internet bubble, and indeed some big ipos.

**thanks** to trevor blackwell, jessica livingston, robert morris, geoff ralston, and harj taggar for reading drafts of this.

04

# ramen profitable

paulgraham.com/ramenprofitable.html

july 2009

now that the term "ramen profitable" has become widespread, i ought to explain precisely what the idea entails.

ramen profitable means a startup makes just enough to pay the founders' living expenses. this is a different form of profitability than startups have traditionally aimed for. traditional profitability means a big bet is finally paying off, whereas the main importance of ramen profitability is that it buys you time. [1]

in the past, a startup would usually become profitable only after raising and spending quite a lot of money. a company making computer hardware might not become profitable for 5 years, during which they spent $50 million. but when they did they might have revenues of $50 million a year. this kind of profitability means the startup has succeeded.

ramen profitability is the other extreme: a startup that becomes profitable after 2 months, even though its revenues are only $3000 a month, because the only employees are a couple 25 year old founders who can live on practically nothing. revenues of $3000 a month do not mean the company has succeeded. but it does share something with the one that's profitable in the traditional way: they don't need to raise money to survive.

ramen profitability is an unfamiliar idea to most people because it only recently became feasible. it's still not feasible for a lot of startups; it would not be for most biotech startups, for example; but it is for many software startups because they're now so cheap. for many, the only real cost is the founders' living expenses.

the main significance of this type of profitability is that you're no longer at the mercy of investors. if you're still losing money, then eventually you'll either have to raise more or shut down. once you're ramen profitable this painful choice goes away. you can still raise money, but you don't have to do it now.

* * *

the most obvious advantage of not needing money is that you can get better terms. if investors know you need money, they'll sometimes take advantage of you. some may even deliberately stall, because they know that as you run out of money you'll become increasingly pliable.

but there are also three less obvious advantages of ramen profitability. one is that it makes you more attractive to investors. if you're already profitable, on however small a scale, it shows that (a) you can get at least someone to pay you, (b) you're serious about building things people want, and (c) you're disciplined enough to keep expenses low.

this is reassuring to investors, because you've addressed three of their biggest worries. it's common for them to fund companies that have smart founders and a big market, and yet still fail. when these companies fail, it's usually because (a) people wouldn't pay for what they made, e.g. because it was too hard to sell to them, or the market wasn't ready yet, (b) the founders solved the wrong problem, instead of paying attention to what users needed, or (c) the company spent too much and burned through their funding before they started to make money. if you're ramen profitable, you're already avoiding these mistakes.

another advantage of ramen profitability is that it's good for morale. a company tends to feel rather theoretical when you first start it. it's legally a company, but you feel like you're lying when you call it

one. when people start to pay you significant amounts, the company starts to feel real. and your own living expenses are the milestone you feel most, because at that point the future flips state. now survival is the default, instead of dying.

a morale boost on that scale is very valuable in a startup, because the moral weight of running a startup is what makes it hard. startups are still very rare. why don't more people do it? the financial risk? plenty of 25 year olds save nothing anyway. the long hours? plenty of people work just as long hours in regular jobs. what keeps people from starting startups is the fear of having so much responsibility. and this is not an irrational fear: it really is hard to bear. anything that takes some of that weight off you will greatly increase your chances of surviving.

a startup that reaches ramen profitability may be more likely to succeed than not. which is pretty exciting, considering the bimodal distribution of outcomes in startups: you either fail or make a lot of money.

the fourth advantage of ramen profitability is the least obvious but may be the most important. if you don't need to raise money, you don't have to interrupt working on the company to do it.

raising money is terribly distracting. you're lucky if your productivity is a third of what it was before. and it can last for months.

i didn't understand (or rather, remember) precisely why raising money was so distracting till earlier this year. i'd noticed that startups we funded would usually grind to a halt when they switched to raising money, but i didn't remember exactly why till yc raised money itself. we had a comparatively easy time of it; the first people i asked said yes; but it took months to work out the details, and during that time i got hardly any real work done. why? because i thought about it all the time.

at any given time there tends to be one problem that's the most urgent for a startup. this is what you think about as you fall asleep at night and when you take a shower in the morning. and when you start raising money, that becomes the problem you think about. you only take one shower in the morning, and if you're thinking about investors during it, then you're not thinking about the product.

whereas if you can choose when you raise money, you can pick a time when you're not in the middle of something else, and you can probably also insist that the round close fast. you may even be able to avoid having the round occupy your thoughts, if you don't care whether it closes.

* * *

ramen profitable means no more than the definition implies. it does not, for example, imply that you're "bootstrapping" the startup—that you're never going to take money from investors. empirically that doesn't seem to work very well. few startups succeed without taking investment. maybe as startups get cheaper it will become more common. on the other hand, the money is there, waiting to be invested. if startups need it less, they'll be able to get it on better terms, which will make them more inclined to take it. that will tend to produce an equilibrium. [2]

another thing ramen profitability doesn't imply is joe kraus's idea that you should put your business model in beta when you put your product in beta. he believes you should get people to pay you from the beginning. i think that's too constraining. facebook didn't, and they've done better than most startups. making money right away was not only unnecessary for them, but probably would have been harmful. i do think joe's rule could be useful for many startups, though. when founders seem unfocused, i sometimes suggest they try to get customers to pay them for something, in the hope that this constraint will prod them into action.

the difference between joe's idea and ramen profitability is that a ramen profitable company doesn't have to be making money the way it ultimately will. it just has to be making money. the most famous example is google, which initially made money by licensing search to sites like yahoo.

is there a downside to ramen profitability? probably the biggest danger is that it might turn you into a consulting firm. startups have to be product companies, in the sense of making a single thing that everyone uses. the defining quality of startups is that they grow fast, and consulting just can't scale the way a product can. [3] but it's pretty easy to make $3000 a month consulting; in fact, that would be a low rate for contract programming. so there could be a temptation to slide into consulting, and telling yourselves you're a ramen profitable startup, when in fact you're not a startup at all.

it's ok to do a little consulting-type work at first. startups usually have to do something weird at first. but remember that ramen profitability is not the destination. a startup's destination is to grow really big; ramen profitability is a trick for not dying en route.

---

## notes

1 the "ramen" in "ramen profitable" refers to instant ramen, which is just about the cheapest food available.

please do not take the term literally. living on instant ramen would be very unhealthy. rice and beans are a better source of food. start by investing in a rice cooker, if you don't have one.

rice and beans for 2n

```
 olive oil or butter n yellow onions other fresh vegetables; experiment 3n cloves garlic n
12-oz cans white, kidney, or black beans n cubes knorr beef or vegetable bouillon n teaspoons
freshly ground black pepper 3n teaspoons ground cumin n cups dry rice, preferably brown
```

put rice in rice cooker. add water as specified on rice package. (default: 2 cups water per cup of rice.) turn on rice cooker and forget about it.

chop onions and other vegetables and fry in oil, over fairly low heat, till onions are glassy. put in chopped garlic, pepper, cumin, and a little more fat, and stir. keep heat low. cook another 2 or 3 minutes, then add beans (don't drain the beans), and stir. throw in the bouillon cube(s), cover, and cook on lowish heat for at least 10 minutes more. stir vigilantly to avoid sticking.

if you want to save money, buy beans in giant cans from discount stores. spices are also much cheaper when bought in bulk. if there's an indian grocery store near you, they'll have big bags of cumin for the same price as the little jars in supermarkets.

2 there's a good chance that a shift in power from investors to founders would actually increase the size of the venture business. i think investors currently err too far on the side of being harsh to founders. if they were forced to stop, the whole venture business would work better, and you might see something like the increase in trade you always see when restrictive laws are removed.

investors are one of the biggest sources of pain for founders; if they stopped causing so much pain, it would be better to be a founder; and if it were better to be a founder, more people would do it.

3 it's conceivable that a startup could grow big by transforming consulting into a form that would scale. but if they did that they'd really be a product company.

**thanks** to jessica livingston for reading drafts of this.

05

# default alive or default dead?

october 2015
paulgraham.com/aord.html

when i talk to a startup that's been operating for more than 8 or 9 months, the first thing i want to know is almost always the same. assuming their expenses remain constant and their revenue growth is what it has been over the last several months, do they make it to profitability on the money they have left? or to put it more dramatically, by default do they live or die?

the startling thing is how often the founders themselves don't know. half the founders i talk to don't know whether they're default alive or default dead.

if you're among that number, trevor blackwell has made a handy calculator you can use to find out.

the reason i want to know first whether a startup is default alive or default dead is that the rest of the conversation depends on the answer. if the company is default alive, we can talk about ambitious new things they could do. if it's default dead, we probably need to talk about how to save it. we know the current trajectory ends badly. how can they get off that trajectory?

why do so few founders know whether they're default alive or default dead? mainly, i think, because they're not used to asking that. it's not a question that makes sense to ask early on, any more than it makes sense to ask a 3 year old how he plans to support himself. but as the company grows older, the question switches from meaningless to critical. that kind of switch often takes people by surprise.

i propose the following solution: instead of starting to ask too late whether you're default alive or default dead, start asking too early. it's hard to say precisely when the question switches polarity. but it's probably not that dangerous to start worrying too early that you're default dead, whereas it's very dangerous to start worrying too late.

the reason is a phenomenon i wrote about earlier: the fatal pinch. the fatal pinch is default dead + slow growth + not enough time to fix it. and the way founders end up in it is by not realizing that's where they're headed.

there is another reason founders don't ask themselves whether they're default alive or default dead: they assume it will be easy to raise more money. but that assumption is often false, and worse still, the more you depend on it, the falser it becomes.

maybe it will help to separate facts from hopes. instead of thinking of the future with vague optimism, explicitly separate the components. say "we're default dead, but we're counting on investors to save us." maybe as you say that, it will set off the same alarms in your head that it does in mine. and if you set off the alarms sufficiently early, you may be able to avoid the fatal pinch.

it would be safe to be default dead if you could count on investors saving you. as a rule their interest is a function of growth. if you have steep revenue growth, say over 5x a year, you can start to count on investors being interested even if you're not profitable. [1] but investors are so fickle that you can never do more than start to count on them. sometimes something about your business will spook investors even if your growth is great. so no matter how good your growth is, you can never safely treat fundraising as more than a plan a. you should always have a plan b as well: you should know (as in write down) precisely what you'll need to do to survive if you can't raise more money, and precisely when you'll have to switch to plan b if plan a isn't working.

in any case, growing fast versus operating cheaply is far from the sharp dichotomy many founders assume it to be. in practice there is surprisingly little connection between how much a startup spends and how fast it grows. when a startup grows fast, it's usually because the product hits a nerve, in the sense of hitting some big need straight on. when a startup spends a lot, it's usually because the product is expensive to develop or sell, or simply because they're wasteful.

if you're paying attention, you'll be asking at this point not just how to avoid the fatal pinch, but how to avoid being default dead. that one is easy: don't hire too fast. hiring too fast is by far the biggest killer of startups that raise money. [2]

founders tell themselves they need to hire in order to grow. but most err on the side of overestimating this need rather than underestimating it. why? partly because there's so much work to do. naive founders think that if they can just hire enough people, it will all get done. partly because successful startups have lots of employees, so it seems like that's what one does in order to be successful. in fact the large staffs of successful startups are probably more the effect of growth than the cause. and partly because when founders have slow growth they don't want to face what is usually the real reason: the product is not appealing enough.

plus founders who've just raised money are often encouraged to overhire by the vcs who funded them. kill-or-cure strategies are optimal for vcs because they're protected by the portfolio effect. vcs want to blow you up, in one sense of the phrase or the other. but as a founder your incentives are different. you want above all to survive. [3]

here's a common way startups die. they make something moderately appealing and have decent initial growth. they raise their first round fairly easily, because the founders seem smart and the idea sounds plausible. but because the product is only moderately appealing, growth is ok but not great. the founders convince themselves that hiring a bunch of people is the way to boost growth. their investors agree. but (because the product is only moderately appealing) the growth never comes. now they're rapidly running out of runway. they hope further investment will save them. but because they have high expenses and slow growth, they're now unappealing to investors. they're unable to raise more, and the company dies.

what the company should have done is address the fundamental problem: that the product is only moderately appealing. hiring people is rarely the way to fix that. more often than not it makes it harder. at this early stage, the product needs to evolve more than to be "built out," and that's usually easier with fewer people. [4]

asking whether you're default alive or default dead may save you from this. maybe the alarm bells it sets off will counteract the forces that push you to overhire. instead you'll be compelled to seek growth in other ways. for example, by doing things that don't scale, or by redesigning the product in the way only founders can. and for many if not most startups, these paths to growth will be the ones that actually work.

airbnb waited 4 months after raising money at the end of y  combinator before they hired their first employee. in the meantime the founders were terribly overworked. but they were overworked evolving airbnb into the astonishingly successful organism it is now.

---

## notes

1 steep usage growth will also interest investors. revenue will ultimately be a constant multiple of usage, so x% usage growth predicts x% revenue growth. but in practice investors discount merely predicted revenue, so if you're measuring usage you need a higher growth rate to impress investors.

2 startups that don't raise money are saved from hiring too fast because they can't afford to. but that doesn't mean you should avoid raising money in order to avoid this problem, any more than that total abstinence is

the only way to avoid becoming an alcoholic.

3 i would not be surprised if vcs' tendency to push founders to overhire is not even in their own interest. they don't know how many of the companies that get killed by overspending might have done well if they'd survived. my guess is a significant number.

4 after reading a draft, sam altman wrote:

"i think you should make the hiring point more strongly. i think it's roughly correct to say that yc's most successful companies have never been the fastest to hire, and one of the marks of a great founder is being able to resist this urge."

paul buchheit adds:

"a related problem that i see a lot is premature scaling—founders take a small business that isn't really working (bad unit economics, typically) and then scale it up because they want impressive growth numbers. this is similar to over-hiring in that it makes the business much harder to fix once it's big, plus they are bleeding cash really fast."

**thanks** to sam altman, paul buchheit, joe gebbia, jessica livingston, and geoff ralston for reading drafts of this.

06

# the fatal pinch

december 2014
paulgraham.com/pinch.html

many startups go through a point a few months before they die where although they have a significant amount of money in the bank, they're also losing a lot each month, and revenue growth is either nonexistent or mediocre. the company has, say, 6 months of runway. or to put it more brutally, 6 months before they're out of business. they expect to avoid that by raising more from investors. [1]

that last sentence is the fatal one.

there may be nothing founders are so prone to delude themselves about as how interested investors will be in giving them additional funding. it's hard to convince investors the first time too, but founders expect that. what bites them the second time is a confluence of three forces:

1. the company is spending more now than it did the first time it raised money.
2. investors have much higher standards for companies that have already raised money.
3. the company is now starting to read as a failure. the first time it raised money, it was neither a success nor a failure; it was too early to ask. now it's possible to ask that question, and the default answer is failure, because at this point that is the default outcome.

i'm going to call the situation i described in the first paragraph "the fatal pinch." i try to resist coining phrases, but making up a name for this situation may snap founders into realizing when they're in it.

one of the things that makes the fatal pinch so dangerous is that it's self-reinforcing. founders overestimate their chances of raising more money, and so are slack about reaching profitability, which further decreases their chances of raising money.

now that you know about the fatal pinch, how do you avoid it? y  combinator tells founders who raise money to act as if it's the last they'll ever get. because the self-reinforcing nature of this situation works the other way too: the less you need further investment, the easier it is to get.

what do you do if you're already in the fatal pinch? the first step is to re-evaluate the probability of raising more money. i will now, by an amazing feat of clairvoyance, do this for you: the probability is zero. [2]

three options remain: you can shut down the company, you can increase how much you make, and you can decrease how much you spend.

you should shut down the company if you're certain it will fail no matter what you do. then at least you can give back the money you have left, and save yourself however many months you would have spent riding it down.

companies rarely *have* to fail though. what i'm really doing here is giving you the option of admitting you've already given up.

if you don't want to shut down the company, that leaves increasing revenues and decreasing expenses. in most startups, expenses = people, and decreasing expenses = firing people. [3] deciding to fire people is usually hard, but there's one case in which it shouldn't be: when there are people you already know you should fire but you're in denial about it. if so, now's the time.

if that makes you profitable, or will enable you to make it to profitability on the money you have left, you've avoided the immediate danger.

otherwise you have three options: you either have to fire good people, get some or all of the employees to take less salary for a while, or increase revenues.

getting people to take less salary is a weak solution that will only work when the problem isn't too bad. if your current trajectory won't quite get you to profitability but you can get over the threshold by cutting salaries a little, you might be able to make the case to everyone for doing it. otherwise you're probably just postponing the problem, and that will be obvious to the people whose salaries you're proposing to cut. [4]

which leaves two options, firing good people and making more money. while trying to balance them, keep in mind the eventual goal: to be a successful product company in the sense of having a single thing lots of people use.

you should lean more toward firing people if the source of your trouble is overhiring. if you went out and hired 15 people before you even knew what you were building, you've created a broken company. you need to figure out what you're building, and it will probably be easier to do that with a handful of people than 15. plus those 15 people might not even be the ones you need for whatever you end up building. so the solution may be to shrink and then figure out what direction to grow in. after all, you're not doing those 15 people any favors if you fly the company into ground with them aboard. they'll all lose their jobs eventually, along with all the time they expended on this doomed company.

whereas if you only have a handful of people, it may be better to focus on trying to make more money. it may seem facile to suggest a startup make more money, as if that could be done for the asking. usually a startup is already trying as hard as it can to sell whatever it sells. what i'm suggesting here is not so much to try harder to make money but to try to make money in a different way. for example, if you have only one person selling while the rest are writing code, consider having everyone work on selling. what good will more code do you when you're out of business? if you have to write code to close a certain deal, go ahead; that follows from everyone working on selling. but only work on whatever will get you the most revenue the soonest.

another way to make money differently is to sell different things, and in particular to do more consultingish work. i say consultingish because there is a long slippery slope from making products to pure consulting, and you don't have to go far down it before you start to offer something really attractive to customers. although your product may not be very appealing yet, if you're a startup your programmers will often be way better than the ones your customers have. or you may have expertise in some new field they don't understand. so if you change your sales conversations just a little from "do you want to buy our product?" to "what do you need that you'd pay a lot for?" you may find it's suddenly a lot easier to extract money from customers.

be ruthlessly mercenary when you start doing this, though. you're trying to save your company from death here, so make customers pay a lot, quickly. and to the extent you can, try to avoid the worst pitfalls of consulting. the ideal thing might be if you built a precisely defined derivative version of your product for the customer, and it was otherwise a straight product sale. you keep the ip and no billing by the hour.

in the best case, this consultingish work may not be just something you do to survive, but may turn out to be the thing-that-doesn't-scale that defines your company. don't expect it to be, but as you dive into individual users' needs, keep your eyes open for narrow openings that have wide vistas beyond.

there is usually so much demand for custom work that unless you're really incompetent there has to be some point down the slope of consulting at which you can survive. but i didn't use the term slippery slope by accident; customers' insatiable demand for custom work will always be pushing you toward

the bottom. so while you'll probably survive, the problem now becomes to survive with the least damage and distraction.

the good news is, plenty of successful startups have passed through near-death experiences and gone on to flourish. you just have to realize in time that you're near death. and if you're in the fatal pinch, you are.

---

## notes

1 there are a handful of companies that can't reasonably expect to make money for the first year or two, because what they're building takes so long. for these companies substitute "progress" for "revenue growth." you're not one of these companies unless your initial investors agreed in advance that you were. and frankly even these companies wish they weren't, because the illiquidity of "progress" puts them at the mercy of investors.

2 there's a variant of the fatal pinch where your existing investors help you along by promising to invest more. or rather, where you read them as promising to invest more, while they think they're just mentioning the possibility. the way to solve this problem, if you have 8 months of runway or less, is to try to get the money right now. then you'll either get the money, in which case (immediate) problem solved, or at least prevent your investors from helping you to remain in denial about your fundraising prospects.

3 obviously, if you have significant expenses other than salaries that you can eliminate, do it now.

4 unless of course the source of the problem is that you're paying yourselves high salaries. if by cutting the founders' salaries to the minimum you need, you can make it to profitability, you should. but it's a bad sign if you needed to read this to realize that.

**thanks** to sam altman, paul buchheit, jessica livingston, and geoff ralston for reading drafts of this.

07

# the equity equation

july 2007
paulgraham.com/equity.html

an investor wants to give you money for a certain percentage of your startup. should you take it? you're about to hire your first employee. how much stock should you give him?

these are some of the hardest questions founders face. and yet both have the same answer:

1/(1 - n)

whenever you're trading stock in your company for anything, whether it's money or an employee or a deal with another company, the test for whether to do it is the same. you should give up n% of your company if what you trade it for improves your average outcome enough that the (100 - n)% you have left is worth more than the whole company was before.

for example, if an investor wants to buy half your company, how much does that investment have to improve your average outcome for you to break even? obviously it has to double: if you trade half your company for something that more than doubles the company's average outcome, you're net ahead. you have half as big a share of something worth more than twice as much.

in the general case, if n is the fraction of the company you're giving up, the deal is a good one if it makes the company worth more than 1/(1 - n).

for example, suppose y combinator offers to fund you in return for 7% of your company. in this case, n is .07 and 1/(1 - n) is 1.075. so you should take the deal if you believe we can improve your average outcome by more than 7.5%. if we improve your outcome by 10%, you're net ahead, because the remaining .93 you hold is worth .93 x 1.1 = 1.023. [1]

one of the things the equity equation shows us is that, financially at least, taking money from a top vc firm can be a really good deal. greg mcadoo from sequoia recently said at a yc dinner that when sequoia invests alone they like to take about 30% of a company. 1/.7 = 1.43, meaning that deal is worth taking if they can improve your outcome by more than 43%. for the average startup, that would be an extraordinary bargain. it would improve the average startup's prospects by more than 43% just to be able to *say* they were funded by sequoia, even if they never actually got the money.

the reason sequoia is such a good deal is that the percentage of the company they take is artificially low. they don't even try to get market price for their investment; they limit their holdings to leave the founders enough stock to feel the company is still theirs.

the catch is that sequoia gets about 6000 business plans a year and funds about 20 of them, so the odds of getting this great deal are 1 in 300. the companies that make it through are not average startups.

of course, there are other factors to consider in a vc deal. it's never just a straight trade of money for stock. but if it were, taking money from a top firm would generally be a bargain.

you can use the same formula when giving stock to employees, but it works in the other direction. if i is the average outcome for the company with the addition of some new person, then they're worth n such that i = 1/(1 - n). which means n = (i - 1)/i.

for example, suppose you're just two founders and you want to hire an additional hacker who's so good you feel he'll increase the average outcome of the whole company by 20%. n = (1.2 - 1)/1.2 = .167. so you'll break even if you trade 16.7% of the company for him.

that doesn't mean 16.7% is the right amount of stock to give him. stock is not the only cost of hiring someone: there's usually salary and overhead as well. and if the company merely breaks even on the deal, there's no reason to do it.

i think to translate salary and overhead into stock you should multiply the annual rate by about 1.5. most startups grow fast or die; if you die you don't have to pay the guy, and if you grow fast you'll be paying next year's salary out of next year's valuation, which should be 3x this year's. if your valuation grows 3x a year, the total cost in stock of a new hire's salary and overhead is 1.5 years' cost at the present valuation.[2]

how much of an additional margin should the company need as the "activation energy" for the deal? since this is in effect the company's profit on a hire, the market will determine that: if you're a hot opportunity, you can charge more.

let's run through an example. suppose the company wants to make a "profit" of 50% on the new hire mentioned above. so subtract a third from 16.7% and we have 11.1% as his "retail" price. suppose further that he's going to cost $60k a year in salary and overhead, x 1.5 = $90k total. if the company's valuation is $2 million, $90k is 4.5%. 11.1% - 4.5% = an offer of 6.6%.

incidentally, notice how important it is for early employees to take little salary. it comes right out of stock that could otherwise be given to them.

obviously there is a great deal of play in these numbers. i'm not claiming that stock grants can now be reduced to a formula. ultimately you always have to guess. but at least know what you're guessing. if you choose a number based on your gut feel, or a table of typical grant sizes supplied by a vc firm, understand what those are estimates of.

and more generally, when you make any decision involving equity, run it through 1/(1 - n) to see if it makes sense. you should always feel richer after trading equity. if the trade didn't increase the value of your remaining shares enough to put you net ahead, you wouldn't have (or shouldn't have) done it.

---

## notes

1 this is why we can't believe anyone would think y combinator was a bad deal. does anyone really think we're so useless that in three months we can't improve a startup's prospects by 7.5%?

2 the obvious choice for your present valuation is the post-money valuation of your last funding round. this probably undervalues the company, though, because (a) unless your last round just happened, the company is presumably worth more, and (b) the valuation of an early funding round usually reflects some other contribution by the investors.

**thanks** to sam altman, trevor blackwell, paul buchheit, hutch fishman, david hornik, paul kedrosky, jessica livingston, gary sabot, and joshua schachter for reading drafts of this.

08

# investor herd dynamics

paulgraham.com/herd.html

august 2013

the biggest component in most investors' opinion of you is the opinion of other investors. which is of course a recipe for exponential growth. when one investor wants to invest in you, that makes other investors want to, which makes others want to, and so on.

sometimes inexperienced founders mistakenly conclude that manipulating these forces is the essence of fundraising. they hear stories about stampedes to invest in successful startups, and think it's therefore the mark of a successful startup to have this happen. but actually the two are not that highly correlated. lots of startups that cause stampedes end up flaming out (in extreme cases, partly as a result of the stampede), and lots of very successful startups were only moderately popular with investors the first time they raised money.

so the point of this essay is not to explain how to create a stampede, but merely to explain the forces that generate them. these forces are always at work to some degree in fundraising, and they can cause surprising situations. if you understand them, you can at least avoid being surprised.

one reason investors like you more when other investors like you is that you actually become a better investment. raising money decreases the risk of failure. indeed, although investors hate it, you are for this reason justified in raising your valuation for later investors. the investors who invested when you had no money were taking more risk, and are entitled to higher returns. plus a company that has raised money is literally more valuable. after you raise the first million dollars, the company is at least a million dollars more valuable, because it's the same company as before, plus it has a million dollars in the bank. [1]

beware, though, because later investors so hate to have the price raised on them that they resist even this self-evident reasoning. only raise the price on an investor you're comfortable with losing, because some will angrily refuse. [2]

the second reason investors like you more when you've had some success at fundraising is that it makes you more confident, and an investors' opinion of you is the foundation of their opinion of your company. founders are often surprised how quickly investors seem to know when they start to succeed at raising money. and while there are in fact lots of ways for such information to spread among investors, the main vector is probably the founders themselves. though they're often clueless about technology, most investors are pretty good at reading people. when fundraising is going well, investors are quick to sense it in your increased confidence. (this is one case where the average founder's inability to remain poker-faced works to your advantage.)

but frankly the most important reason investors like you more when you've started to raise money is that they're bad at judging startups. judging startups is hard even for the best investors. the mediocre ones might as well be flipping coins. so when mediocre investors see that lots of other people want to invest in you, they assume there must be a reason. this leads to the phenomenon known in the valley as the "hot deal," where you have more interest from investors than you can handle.

the best investors aren't influenced much by the opinion of other investors. it would only dilute their own judgment to average it together with other people's. but they are indirectly influenced in the practical sense that interest from other investors imposes a deadline. this is the fourth way in which offers beget offers. if you start to get far along the track toward an offer with one firm, it will sometimes provoke other firms, even good ones, to make up their minds, lest they lose the deal.

unless you're a wizard at negotiation (and if you're not sure, you're not) be very careful about exaggerating this to push a good investor to decide. founders try this sort of thing all the time, and investors are very sensitive to it. if anything oversensitive. but you're safe so long as you're telling the truth. if you're getting far along with investor b, but you'd rather raise money from investor a, you can tell investor a that this is happening. there's no manipulation in that. you're genuinely in a bind, because you really would rather raise money from a, but you can't safely reject an offer from b when it's still uncertain what a will decide.

do not, however, tell a who b is. vcs will sometimes ask which other vcs you're talking to, but you should never tell them. angels you can sometimes tell about other angels, because angels cooperate more with one another. but if vcs ask, just point out that they wouldn't want you telling other firms about your conversations, and you feel obliged to do the same for any firm you talk to. if they push you, point out that you're inexperienced at fundraising — which is always a safe card to play — and you feel you have to be extra cautious. [3]

while few startups will experience a stampede of interest, almost all will at least initially experience the other side of this phenomenon, where the herd remains clumped together at a distance. the fact that investors are so much influenced by other investors' opinions means you always start out in something of a hole. so don't be demoralized by how hard it is to get the first commitment, because much of the difficulty comes from this external force. the second will be easier.

---

## notes

1 an accountant might say that a company that has raised a million dollars is no richer if it's convertible debt, but in practice money raised as convertible debt is little different from money raised in an equity round.

2 founders are often surprised by this, but investors can get very emotional. or rather indignant; that's the main emotion i've observed; but it is very common, to the point where it sometimes causes investors to act against their own interests. i know of one investor who invested in a startup at a $15 million valuation cap. earlier he'd had an opportunity to invest at a $5 million cap, but he refused because a friend who invested earlier had been able to invest at a $3 million cap.

3 if an investor pushes you hard to tell them about your conversations with other investors, is this someone you want as an investor?

**thanks** to paul buchheit, jessica livingston, geoff ralston, and garry tan for reading drafts of this.

09

# high resolution fundraising

paulgraham.com/hiresfund.html

september 2010

the reason startups have been using more convertible notes in angel rounds is that they make deals close faster. by making it easier for startups to give different prices to different investors, they help them break the sort of deadlock that happens when investors all wait to see who else is going to invest.

by far the biggest influence on investors' opinions of a startup is the opinion of other investors. there are very, very few who simply decide for themselves. any startup founder can tell you the most common question they hear from investors is not about the founders or the product, but "who else is investing?"

that tends to produce deadlocks. raising an old-fashioned fixed-size equity round can take weeks, because all the angels sit around waiting for the others to commit, like competitors in a bicycle sprint who deliberately ride slowly at the start so they can follow whoever breaks first.

convertible notes let startups beat such deadlocks by rewarding investors willing to move first with lower (effective) valuations. which they deserve because they're taking more risk. it's much safer to invest in a startup ron conway has already invested in; someone who comes after him should pay a higher price.

the reason convertible notes allow more flexibility in price is that valuation caps aren't actual valuations, and notes are cheap and easy to do. so you can do high-resolution fundraising: if you wanted you could have a separate note with a different cap for each investor.

that cap need not simply rise monotonically. a startup could also give better deals to investors they expected to help them most. the point is simply that different investors, whether because of the help they offer or their willingness to commit, have different values for startups, and their terms should reflect that.

different terms for different investors is clearly the way of the future. markets always evolve toward higher resolution. you may not need to use convertible notes to do it. with sufficiently lightweight standardized equity terms (and some changes in investors' and lawyers' expectations about equity rounds) you might be able to do the same thing with equity instead of debt. either would be fine with startups, so long as they can easily change their valuation.

deadlocks weren't the only problem with fixed-size equity rounds. another was that startups had to decide in advance how much to raise. i think it's a mistake for a startup to fix upon a specific number. if investors are easily convinced, the startup should raise more now, and if investors are skeptical, the startup should take a smaller amount and use that to get the company to the point where it's more convincing.

it's just not reasonable to expect startups to pick an optimal round size in advance, because that depends on the reactions of investors, and those are impossible to predict.

fixed-size, multi-investor angel rounds are such a bad idea for startups that one wonders why things were ever done that way. one possibility is that this custom reflects the way investors like to collude when they can get away with it. but i think the actual explanation is less sinister. i think angels (and their lawyers) organized rounds this way in unthinking imitation of vc series a rounds. in a series a, a fixed-size equity round with a lead makes sense, because there is usually just one big investor, who is unequivocally the lead. fixed-size series a rounds already are high res. but the more investors you have in a round, the less sense it makes for everyone to get the same price.

the most interesting question here may be what high res fundraising will do to the world of investors. bolder investors will now get rewarded with lower prices. but more important, in a hits-driven business, is that they'll be able to get into the deals they want. whereas the "who else is investing?" type of investors will not only pay higher prices, but may not be able to get into the best deals at all.

**thanks** to immad akhund, sam altman, john bautista, pete koomen, jessica livingston, dan siroker, harj taggar, and fred wilson for reading drafts of this.

10

# how to convince investors

paulgraham.com/convince.html

august 2013

when people hurt themselves lifting heavy things, it's usually because they try to lift with their back. the right way to lift heavy things is to let your legs do the work. inexperienced founders make the same mistake when trying to convince investors. they try to convince with their pitch. most would be better off if they let their startup do the work — if they started by understanding why their startup is worth investing in, then simply explained this well to investors.

investors are looking for startups that will be very successful. but that test is not as simple as it sounds. in startups, as in a lot of other domains, the distribution of outcomes follows a power law, but in startups the curve is startlingly steep. the big successes are so big they dwarf the rest. and since there are only a handful each year (the conventional wisdom is 15), investors treat "big success" as if it were binary. most are interested in you if you seem like you have a chance, however small, of being one of the 15 big successes, and otherwise not. [1]

(there are a handful of angels who'd be interested in a company with a high probability of being moderately successful. but angel investors like big successes too.)

how do you seem like you'll be one of the big successes? you need three things: formidable founders, a promising market, and (usually) some evidence of success so far.

**formidable**

the most important ingredient is formidable founders. most investors decide in the first few minutes whether you seem like a winner or a loser, and once their opinion is set it's hard to change. [2] every startup has reasons both to invest and not to invest. if investors think you're a winner they focus on the former, and if not they focus on the latter. for example, it might be a rich market, but with a slow sales cycle. if investors are impressed with you as founders, they say they want to invest because it's a rich market, and if not, they say they can't invest because of the slow sales cycle.

they're not necessarily trying to mislead you. most investors are genuinely unclear in their own minds why they like or dislike startups. if you seem like a winner, they'll like your idea more. but don't be too smug about this weakness of theirs, because you have it too; almost everyone does.

there is a role for ideas of course. they're fuel for the fire that starts with liking the founders. once investors like you, you'll see them reaching for ideas: they'll be saying "yes, and you could also do x." (whereas when they don't like you, they'll be saying "but what about y?")

but the foundation of convincing investors is to seem formidable, and since this isn't a word most people use in conversation much, i should explain what it means. a formidable person is one who seems like they'll get what they want, regardless of whatever obstacles are in the way. formidable is close to confident, except that someone could be confident and mistaken. formidable is roughly justifiably confident.

there are a handful of people who are really good at seeming formidable — some because they actually are very formidable and just let it show, and others because they are more or less con artists. [3] but most founders, including many who will go on to start very successful companies, are not that good at seeming formidable the first time they try fundraising. what should they do? [4]

what they should not do is try to imitate the swagger of more experienced founders. investors are not always that good at judging technology, but they're good at judging confidence. if you try to act like

something you're not, you'll just end up in an uncanny valley. you'll depart from sincere, but never arrive at convincing.

**truth**

the way to seem most formidable as an inexperienced founder is to stick to the truth. how formidable you seem isn't a constant. it varies depending on what you're saying. most people can seem confident when they're saying "one plus one is two," because they know it's true. the most diffident person would be puzzled and even slightly contemptuous if they told a vc "one plus one is two" and the vc reacted with skepticism. the magic ability of people who are good at seeming formidable is that they can do this with the sentence "we're going to make a billion dollars a year." but you can do the same, if not with that sentence with some fairly impressive ones, so long as you convince yourself first.

that's the secret. convince yourself that your startup is worth investing in, and then when you explain this to investors they'll believe you. and by convince yourself, i don't mean play mind games with yourself to boost your confidence. i mean truly evaluate whether your startup is worth investing in. if it isn't, don't try to raise money. [5] but if it is, you'll be telling the truth when you tell investors it's worth investing in, and they'll sense that. you don't have to be a smooth presenter if you understand something well and tell the truth about it.

to evaluate whether your startup is worth investing in, you have to be a domain expert. if you're not a domain expert, you can be as convinced as you like about your idea, and it will seem to investors no more than an instance of the dunning-kruger effect. which in fact it will usually be. and investors can tell fairly quickly whether you're a domain expert by how well you answer their questions. know everything about your market. [6]

why do founders persist in trying to convince investors of things they're not convinced of themselves? partly because we've all been trained to.

when my friends robert morris and trevor blackwell were in grad school, one of their fellow students was on the receiving end of a question from their faculty advisor that we still quote today. when the unfortunate fellow got to his last slide, the professor burst out:

> which one of these conclusions do you actually believe?

one of the artifacts of the way schools are organized is that we all get trained to talk even when we have nothing to say. if you have a ten page paper due, then ten pages you must write, even if you only have one page of ideas. even if you have no ideas. you have to produce something. and all too many startups go into fundraising in the same spirit. when they think it's time to raise money, they try gamely to make the best case they can for their startup. most never think of pausing beforehand to ask whether what they're saying is actually convincing, because they've all been trained to treat the need to present as a given — as an area of fixed size, over which however much truth they have must needs be spread, however thinly.

the time to raise money is not when you need it, or when you reach some artificial deadline like a demo day. it's when you can convince investors, and not before. [7]

and unless you're a good con artist, you'll never convince investors if you're not convinced yourself. they're far better at detecting bullshit than you are at producing it, even if you're producing it unknowingly. if you try to convince investors before you've convinced yourself, you'll be wasting both your time.

but pausing first to convince yourself will do more than save you from wasting your time. it will force you to organize your thoughts. to convince yourself that your startup is worth investing in, you'll have to figure out why it's worth investing in. and if you can do that you'll end up with more than added confidence. you'll also have a provisional roadmap of how to succeed.

**market**

notice i've been careful to talk about whether a startup is worth investing in, rather than whether it's going to succeed. no one knows whether a startup is going to succeed. and it's a good thing for investors that this is so, because if you could know in advance whether a startup would succeed, the stock price would already be the future price, and there would be no room for investors to make money. startup investors know that every investment is a bet, and against pretty long odds.

so to prove you're worth investing in, you don't have to prove you're going to succeed, just that you're a sufficiently good bet. what makes a startup a sufficiently good bet? in addition to formidable founders, you need a plausible path to owning a big piece of a big market. founders think of startups as ideas, but investors think of them as markets. if there are x number of customers who'd pay an average of $y per year for what you're making, then the total addressable market, or tam, of your company is $xy. investors don't expect you to collect all that money, but it's an upper bound on how big you can get.

your target market has to be big, and it also has to be capturable by you. but the market doesn't have to be big yet, nor do you necessarily have to be in it yet. indeed, it's often better to start in a small market that will either turn into a big one or from which you can move into a big one. there just has to be some plausible sequence of hops that leads to dominating a big market a few years down the line.

the standard of plausibility varies dramatically depending on the age of the startup. a three month old company at demo day only needs to be a promising experiment that's worth funding to see how it turns out. whereas a two year old company raising a series a round needs to be able to show the experiment worked. [8]

but every company that gets really big is "lucky" in the sense that their growth is due mostly to some external wave they're riding, so to make a convincing case for becoming huge, you have to identify some specific trend you'll benefit from. usually you can find this by asking "why now?" if this is such a great idea, why hasn't someone else already done it? ideally the answer is that it only recently became a good idea, because something changed, and no one else has noticed yet.

microsoft for example was not going to grow huge selling basic interpreters. but by starting there they were perfectly poised to expand up the stack of microcomputer software as microcomputers grew powerful enough to support one. and microcomputers turned out to be a really huge wave, bigger than even the most optimistic observers would have predicted in 1975.

but while microsoft did really well and there is thus a temptation to think they would have seemed a great bet a few months in, they probably didn't. good, but not great. no company, however successful, ever looks more than a pretty good bet a few months in. microcomputers turned out to be a big deal, and microsoft both executed well and got lucky. but it was by no means obvious that this was how things would play out. plenty of companies seem as good a bet a few months in. i don't know about startups in general, but at least half the startups we fund could make as good a case as microsoft could have for being on a path to dominating a large market. and who can reasonably expect more of a startup than that?

**rejection**

if you can make as good a case as microsoft could have, will you convince investors? not always. a lot of vcs would have rejected microsoft. [9] certainly some rejected google. and getting rejected will put you in a slightly awkward position, because as you'll see when you start fundraising, the most common question you'll get from investors will be "who else is investing?" what do you say if you've been fundraising for a while and no one has committed yet? [10]

the people who are really good at acting formidable often solve this problem by giving investors the impression that while no investors have committed yet, several are about to. this is arguably a permissible tactic. it's slightly dickish of investors to care more about who else is investing than any other aspect of your startup, and misleading them about how far along you are with other investors seems the complementary countermove. it's arguably an instance of scamming a scammer. but i don't recommend this approach to most founders, because most founders wouldn't be able to carry it off. this is the single most common lie told to investors, and you have to be really good at lying to tell members of some profession the most common lie they're told.

if you're not a master of negotiation (and perhaps even if you are) the best solution is to tackle the problem head-on, and to explain why investors have turned you down and why they're mistaken. if you know you're on the right track, then you also know why investors were wrong to reject you. experienced investors are well aware that the best ideas are also the scariest. they all know about the vcs who rejected google. if instead of seeming evasive and ashamed about having been turned down (and thereby implicitly agreeing with the verdict) you talk candidly about what scared investors about you, you'll seem more confident, which they like, and you'll probably also do a better job of presenting that aspect of your startup. at the very least, that worry will now be out in the open instead of being a gotcha left to be discovered by the investors you're currently talking to, who will be proud of and thus attached to their discovery. [11]

this strategy will work best with the best investors, who are both hard to bluff and who already believe most other investors are conventional-minded drones doomed always to miss the big outliers. raising money is not like applying to college, where you can assume that if you can get into mit, you can also get into foobar state. because the best investors are much smarter than the rest, and the best startup ideas look initially like bad ideas, it's not uncommon for a startup to be rejected by all the vcs except the best ones. that's what happened to dropbox. y combinator started in boston, and for the first 3 years we ran alternating batches in boston and silicon valley. because boston investors were so few and so timid, we used to ship boston batches out for a second demo day in silicon valley. dropbox was part of a boston batch, which means all those boston investors got the first look at dropbox, and none of them closed the deal. yet another backup and syncing thing, they all thought. a couple weeks later, dropbox raised a series a round from sequoia. [12]

**different**

not understanding that investors view investments as bets combines with the ten page paper mentality to prevent founders from even considering the possibility of being certain of what they're saying. they think they're trying to convince investors of something very uncertain — that their startup will be huge — and convincing anyone of something like that must obviously entail some wild feat of salesmanship. but in fact when you raise money you're trying to convince investors of something so much less speculative — whether the company has all the elements of a good bet — that you can approach the problem in a qualitatively different way. you can convince yourself, then convince them.

and when you convince them, use the same matter-of-fact language you used to convince yourself. you wouldn't use vague, grandiose marketing-speak among yourselves. don't use it with investors either. it

not only doesn't work on them, but seems a mark of incompetence. just be concise. many investors explicitly use that as a test, reasoning (correctly) that if you can't explain your plans concisely, you don't really understand them. but even investors who don't have a rule about this will be bored and frustrated by unclear explanations. [13]

so here's the recipe for impressing investors when you're not already good at seeming formidable:

1. make something worth investing in.
2. understand why it's worth investing in.
3. explain that clearly to investors.

if you're saying something you know is true, you'll seem confident when you're saying it. conversely, never let pitching draw you into bullshitting. as long as you stay on the territory of truth, you're strong. make the truth good, then just tell it.

---

## notes

1 there's no reason to believe this number is a constant. in fact it's our explicit goal at y combinator to increase it, by encouraging people to start startups who otherwise wouldn't have.

2 or more precisely, investors decide whether you're a loser or possibly a winner. if you seem like a winner, they may then, depending on how much you're raising, have several more meetings with you to test whether that initial impression holds up.

but if you seem like a loser they're done, at least for the next year or so. and when they decide you're a loser they usually decide in way less than the 50 minutes they may have allotted for the first meeting. which explains the astonished stories one always hears about vc inattentiveness. how could these people make investment decisions well when they're checking their messages during startups' presentations? the solution to that mystery is that they've already made the decision.

3 the two are not mutually exclusive. there are people who are both genuinely formidable, and also really good at acting that way.

4 how can people who will go on to create giant companies not seem formidable early on? i think the main reason is that their experience so far has trained them to keep their wings folded, as it were. family, school, and jobs encourage cooperation, not conquest. and it's just as well they do, because even being genghis khan is probably 99% cooperation. but the result is that most people emerge from the tube of their upbringing in their early twenties compressed into the shape of the tube. some find they have wings and start to spread them. but this takes a few years. in the beginning even they don't know yet what they're capable of.

5 in fact, change what you're doing. you're investing your own time in your startup. if you're not convinced that what you're working on is a sufficiently good bet, why are you even working on that?

6 when investors ask you a question you don't know the answer to, the best response is neither to bluff nor give up, but instead to explain how you'd figure out the answer. if you can work out a preliminary answer on the spot, so much the better, but explain that's what you're doing.

7 at yc we try to ensure startups are ready to raise money on demo day by encouraging them to ignore investors and instead focus on their companies till about a week before. that way most reach the stage where they're sufficiently convincing well before demo day. but not all do, so we also give any startup that wants to the option of deferring to a later demo day.

8 founders are often surprised by how much harder it is to raise the next round. there is a qualitative difference in investors' attitudes. it's like the difference between being judged as a kid and as an adult. the next time you

raise money, it's not enough to be promising. you have to be delivering results.

so although it works well to show growth graphs at either stage, investors treat them differently. at three months, a growth graph is mostly evidence that the founders are effective. at two years, it has to be evidence of a promising market and a company tuned to exploit it.

9 by this i mean that if the present day equivalent of the 3 month old microsoft presented at a demo day, there would be investors who turned them down. microsoft itself didn't raise outside money, and indeed the venture business barely existed when they got started in 1975.

10 the best investors rarely care who else is investing, but mediocre investors almost all do. so you can use this question as a test of investor quality.

11 to use this technique, you'll have to find out why investors who rejected you did so, or at least what they claim was the reason. that may require asking, because investors don't always volunteer a lot of detail. make it clear when you ask that you're not trying to dispute their decision — just that if there is some weakness in your plans, you need to know about it. you won't always get a real reason out of them, but you should at least try.

12 dropbox wasn't rejected by all the east coast vcs. there was one firm that wanted to invest but tried to lowball them.

13 alfred lin points out that it's doubly important for the explanation of a startup to be clear and concise, because it has to convince at one remove: it has to work not just on the partner you talk to, but when that partner re-tells it to colleagues.

we consciously optimize for this at yc. when we work with founders create a demo day pitch, the last step is to imagine how an investor would sell it to colleagues.

**thanks** to marc andreessen, sam altman, patrick collison, ron conway, chris dixon, alfred lin, ben horowitz, steve huffman, jessica livingston, greg mcadoo, andrew mason, geoff ralston, yuri sagalov, emmett shear, rajat suri, garry tan, albert wenger, fred wilson, and qasar younis for reading drafts of this.

11

# how to raise money

paulgraham.com/fr.html

september 2013

most startups that raise money do it more than once. a typical trajectory might be (1) to get started with a few tens of thousands from something like y combinator or individual angels, then (2) raise a few hundred thousand to a few million to build the company, and then (3) once the company is clearly succeeding, raise one or more later rounds to accelerate growth.

reality can be messier. some companies raise money twice in phase 2. others skip phase 1 and go straight to phase 2. and at y combinator we get an increasing number of companies that have already raised amounts in the hundreds of thousands. but the three phase path is at least the one about which individual startups' paths oscillate.

this essay focuses on phase 2 fundraising. that's the type the startups we fund are doing on demo day, and this essay is the advice we give them.

**forces**

fundraising is hard in both senses: hard like lifting a heavy weight, and hard like solving a puzzle. it's hard like lifting a weight because it's intrinsically hard to convince people to part with large sums of money. that problem is irreducible; it should be hard. but much of the other kind of difficulty can be eliminated. fundraising only seems a puzzle because it's an alien world to most founders, and i hope to fix that by supplying a map through it.

to founders, the behavior of investors is often opaque — partly because their motivations are obscure, but partly because they deliberately mislead you. and the misleading ways of investors combine horribly with the wishful thinking of inexperienced founders. at yc we're always warning founders about this danger, and investors are probably more circumspect with yc startups than with other companies they talk to, and even so we witness a constant series of explosions as these two volatile components combine. [1]

if you're an inexperienced founder, the only way to survive is by imposing external constraints on yourself. you can't trust your intuitions. i'm going to give you a set of rules here that will get you through this process if anything will. at certain moments you'll be tempted to ignore them. so rule number zero is: these rules exist for a reason. you wouldn't need a rule to keep you going in one direction if there weren't powerful forces pushing you in another.

the ultimate source of the forces acting on you are the forces acting on investors. investors are pinched between two kinds of fear: fear of investing in startups that fizzle, and fear of missing out on startups that take off. the cause of all this fear is the very thing that makes startups such attractive investments: the successful ones grow very fast. but that fast growth means investors can't wait around. if you wait till a startup is obviously a success, it's too late. to get the really high returns, you have to invest in startups when it's still unclear how they'll do. but that in turn makes investors nervous they're about to invest in a flop. as indeed they often are.

what investors would like to do, if they could, is wait. when a startup is only a few months old, every week that passes gives you significantly more information about them. but if you wait too long, other investors might take the deal away from you. and of course the other investors are all subject to the same forces. so what tends to happen is that they all wait as long as they can, then when some act the rest have to.

**don't raise money unless you want it and it wants you.**

such a high proportion of successful startups raise money that it might seem fundraising is one of the defining qualities of a startup. actually it isn't. rapid growth is what makes a company a startup. most companies in a position to grow rapidly find that (a) taking outside money helps them grow faster, and (b) their growth potential makes it easy to attract such money. it's so common for both (a) and (b) to be true of a successful startup that practically all do raise outside money. but there may be cases where a startup either wouldn't want to grow faster, or outside money wouldn't help them to, and if you're one of them, don't raise money.

the other time not to raise money is when you won't be able to. if you try to raise money before you can convince investors, you'll not only waste your time, but also burn your reputation with those investors.

**be in fundraising mode or not.**

one of the things that surprises founders most about fundraising is how distracting it is. when you start fundraising, everything else grinds to a halt. the problem is not the time fundraising consumes but that it becomes the top idea in your mind. a startup can't endure that level of distraction for long. an early stage startup grows mostly because the founders make it grow, and if the founders look away, growth usually drops sharply.

because fundraising is so distracting, a startup should either be in fundraising mode or not. and when you do decide to raise money, you should focus your whole attention on it so you can get it done quickly and get back to work. [2]

you can take money from investors when you're not in fundraising mode. you just can't expend any attention on it. there are two things that take attention: convincing investors, and negotiating with them. so when you're not in fundraising mode, you should take money from investors only if they require no convincing, and are willing to invest on terms you'll take without negotiation. for example, if a reputable investor is willing to invest on a convertible note, using standard paperwork, that is either uncapped or capped at a good valuation, you can take that without having to think. [3] the terms will be whatever they turn out to be in your next equity round. and "no convincing" means just that: zero time spent meeting with investors or preparing materials for them. if an investor says they're ready to invest, but they need you to come in for one meeting to meet some of the partners, tell them no, if you're not in fundraising mode, because that's fundraising. [4] tell them politely; tell them you're focusing on the company right now, and that you'll get back to them when you're fundraising; but do not get sucked down the slippery slope.

investors will try to lure you into fundraising when you're not. it's great for them if they can, because they can thereby get a shot at you before everyone else. they'll send you emails saying they want to meet to learn more about you. if you get cold-emailed by an associate at a vc firm, you shouldn't meet even if you are in fundraising mode. deals don't happen that way. [5] but even if you get an email from a partner you should try to delay meeting till you're in fundraising mode. they may say they just want to meet and chat, but investors never just want to meet and chat. what if they like you? what if they start to talk about giving you money? will you be able to resist having that conversation? unless you're experienced enough at fundraising to have a casual conversation with investors that stays casual, it's safer to tell them that you'd be happy to later, when you're fundraising, but that right now you need to focus on the company. [6]

companies that are successful at raising money in phase 2 sometimes tack on a few investors after leaving fundraising mode. this is fine; if fundraising went well, you'll be able to do it without spending

time convincing them or negotiating about terms.

**get introductions to investors.**

before you can talk to investors, you have to be introduced to them. if you're presenting at a demo day, you'll be introduced to a whole bunch simultaneously. but even if you are, you should supplement these with intros you collect yourself.

do you have to be introduced? in phase 2, yes. some investors will let you email them a business plan, but you can tell from the way their sites are organized that they don't really want startups to approach them directly.

intros vary greatly in effectiveness. the best type of intro is from a well-known investor who has just invested in you. so when you get an investor to commit, ask them to introduce you to other investors they respect. [7] the next best type of intro is from a founder of a company they've funded. you can also get intros from other people in the startup community, like lawyers and reporters.

there are now sites like angellist, fundersclub, and wefunder that can introduce you to investors. we recommend startups treat them as auxiliary sources of money. raise money first from leads you get yourself. those will on average be better investors. plus you'll have an easier time raising money on these sites once you can say you've already raised some from well-known investors.

**hear no till you hear yes.**

treat investors as saying no till they unequivocally say yes, in the form of a definite offer with no contingencies.

i mentioned earlier that investors prefer to wait if they can. what's particularly dangerous for founders is the way they wait. essentially, they lead you on. they seem like they're about to invest right up till the moment they say no. if they even say no. some of the worse ones never actually do say no; they just stop replying to your emails. they hope that way to get a free option on investing. if they decide later that they want to invest — usually because they've heard you're a hot deal — they can pretend they just got distracted and then restart the conversation as if they'd been about to. [8]

that's not the worst thing investors will do. some will use language that makes it sound as if they're committing, but which doesn't actually commit them. and wishful thinking founders are happy to meet them half way. [9]

fortunately, the next rule is a tactic for neutralizing this behavior. but to work it depends on you not being tricked by the no that sounds like yes. it's so common for founders to be misled/mistaken about this that we designed a protocol to fix the problem. if you believe an investor has committed, get them to confirm it. if you and they have different views of reality, whether the source of the discrepancy is their sketchiness or your wishful thinking, the prospect of confirming a commitment in writing will flush it out. and till they confirm, regard them as saying no.

**do breadth-first search weighted by expected value.**

when you talk to investors your m.o. should be breadth-first search, weighted by expected value. you should always talk to investors in parallel rather than serially. you can't afford the time it takes to talk to investors serially, plus if you only talk to one investor at a time, they don't have the pressure of other investors to make them act. but you shouldn't pay the same attention to every investor, because some are more promising prospects than others. the optimal solution is to talk to all potential investors in parallel, but give higher priority to the more promising ones. [10]

expected value = how likely an investor is to say yes, multiplied by how good it would be if they did. so for example, an eminent investor who would invest a lot, but will be hard to convince, might have the same expected value as an obscure angel who won't invest much, but will be easy to convince. whereas an obscure angel who will only invest a small amount, and yet needs to meet multiple times before making up his mind, has very low expected value. meet such investors last, if at all. [11]

doing breadth-first search weighted by expected value will save you from investors who never explicitly say no but merely drift away, because you'll drift away from them at the same rate. it protects you from investors who flake in much the same way that a distributed algorithm protects you from processors that fail. if some investor isn't returning your emails, or wants to have lots of meetings but isn't progressing toward making you an offer, you automatically focus less on them. but you have to be disciplined about assigning probabilities. you can't let how much you want an investor influence your estimate of how much they want you.

**know where you stand.**

how do you judge how well you're doing with an investor, when investors habitually seem more positive than they are? by looking at their actions rather than their words. every investor has some track they need to move along from the first conversation to wiring the money, and you should always know what that track consists of, where you are on it, and how fast you're moving forward.

never leave a meeting with an investor without asking what happens next. what more do they need in order to decide? do they need another meeting with you? to talk about what? and how soon? do they need to do something internally, like talk to their partners, or investigate some issue? how long do they expect it to take? don't be too pushy, but know where you stand. if investors are vague or resist answering such questions, assume the worst; investors who are seriously interested in you will usually be happy to talk about what has to happen between now and wiring the money, because they're already running through that in their heads. [12]

if you're experienced at negotiations, you already know how to ask such questions. [13] if you're not, there's a trick you can use in this situation. investors know you're inexperienced at raising money. inexperience there doesn't make you unattractive. being a noob at technology would, if you're starting a technology startup, but not being a noob at fundraising. larry and sergey were noobs at fundraising. so you can just confess that you're inexperienced at this and ask how their process works and where you are in it. [14]

**get the first commitment.**

the biggest factor in most investors' opinions of you is the opinion of other investors. once you start getting investors to commit, it becomes increasingly easy to get more to. but the other side of this coin is that it's often hard to get the first commitment.

getting the first substantial offer can be half the total difficulty of fundraising. what counts as a substantial offer depends on who it's from and how much it is. money from friends and family doesn't usually count, no matter how much. but if you get $50k from a well known vc firm or angel investor, that will usually be enough to set things rolling. [15]

**close committed money.**

it's not a deal till the money's in the bank. i often hear inexperienced founders say things like "we've raised $800,000," only to discover that zero of it is in the bank so far. remember the twin fears that torment investors? the fear of missing out that makes them jump early, and the fear of jumping onto a

turd that results? this is a market where people are exceptionally prone to buyer's remorse. and it's also one that furnishes them plenty of excuses to gratify it. the public markets snap startup investing around like a whip. if the chinese economy blows up tomorrow, all bets are off. but there are lots of surprises for individual startups too, and they tend to be concentrated around fundraising. tomorrow a big competitor could appear, or you could get c&ded, or your cofounder could quit. [16]

even a day's delay can bring news that causes an investor to change their mind. so when someone commits, get the money. knowing where you stand doesn't end when they say they'll invest. after they say yes, know what the timetable is for getting the money, and then babysit that process till it happens. institutional investors have people in charge of wiring money, but you may have to hunt angels down in person to collect a check.

inexperienced investors are the ones most likely to get buyer's remorse. established ones have learned to treat saying yes as like diving off a diving board, and they also have more brand to preserve. but i've heard of cases of even top-tier vc firms welching on deals.

**avoid investors who don't "lead."**

since getting the first offer is most of the difficulty of fundraising, that should be part of your calculation of expected value when you start. you have to estimate not just the probability that an investor will say yes, but the probability that they'd be the *first* to say yes, and the latter is not simply a constant fraction of the former. some investors are known for deciding quickly, and those are extra valuable early on.

conversely, an investor who will only invest once other investors have is worthless initially. and while most investors are influenced by how interested other investors are in you, there are some who have an explicit policy of only investing after other investors have. you can recognize this contemptible subspecies of investor because they often talk about "leads." they say that they don't lead, or that they'll invest once you have a lead. sometimes they even claim to be willing to lead themselves, by which they mean they won't invest till you get $x from other investors. (it's great if by "lead" they mean they'll invest unilaterally, and in addition will help you raise more. what's lame is when they use the term to mean they won't invest unless you can raise more elsewhere.) [17]

where does this term "lead" come from? up till a few years ago, startups raising money in phase 2 would usually raise equity rounds in which several investors invested at the same time using the same paperwork. you'd negotiate the terms with one "lead" investor, and then all the others would sign the same documents and all the money change hands at the closing.

series a rounds still work that way, but things now work differently for most fundraising prior to the series a. now there are rarely actual rounds before the a round, or leads for them. now startups simply raise money from investors one at a time till they feel they have enough.

since there are no longer leads, why do investors use that term? because it's a more legitimate-sounding way of saying what they really mean. all they really mean is that their interest in you is a function of other investors' interest in you. i.e. the spectral signature of all mediocre investors. but when phrased in terms of leads, it sounds like there is something structural and therefore legitimate about their behavior.

when an investor tells you "i want to invest in you, but i don't lead," translate that in your mind to "no, except yes if you turn out to be a hot deal." and since that's the default opinion of any investor about any startup, they've essentially just told you nothing.

when you first start fundraising, the expected value of an investor who won't "lead" is zero, so talk to such investors last if at all.

**have multiple plans.**

many investors will ask how much you're planning to raise. this question makes founders feel they should be planning to raise a specific amount. but in fact you shouldn't. it's a mistake to have fixed plans in an undertaking as unpredictable as fundraising.

so why do investors ask how much you plan to raise? for much the same reasons a salesperson in a store will ask "how much were you planning to spend?" if you walk in looking for a gift for a friend. you probably didn't have a precise amount in mind; you just want to find something good, and if it's inexpensive, so much the better. the salesperson asks you this not because you're supposed to have a plan to spend a specific amount, but so they can show you only things that cost the most you'll pay.

similarly, when investors ask how much you plan to raise, it's not because you're supposed to have a plan. it's to see whether you'd be a suitable recipient for the size of investment they like to make, and also to judge your ambition, reasonableness, and how far you are along with fundraising.

if you're a wizard at fundraising, you can say "we plan to raise a $7 million series a round, and we'll be accepting termsheets next tuesday." i've known a handful of founders who could pull that off without having vcs laugh in their faces. but if you're in the inexperienced but earnest majority, the solution is analogous to the solution i recommend for pitching your startup: do the right thing and then just tell investors what you're doing.

and the right strategy, in fundraising, is to have multiple plans depending on how much you can raise. ideally you should be able to tell investors something like: we can make it to profitability without raising any more money, but if we raise a few hundred thousand we can hire one or two smart friends, and if we raise a couple million, we can hire a whole engineering team, etc.

different plans match different investors. if you're talking to a vc firm that only does series a rounds (though there are few of those left), it would be a waste of time talking about any but your most expensive plan. whereas if you're talking to an angel who invests $20k at a time and you haven't raised any money yet, you probably want to focus on your least expensive plan.

if you're so fortunate as to have to think about the upper limit on what you should raise, a good rule of thumb is to multiply the number of people you want to hire times $15k times 18 months. in most startups, nearly all the costs are a function of the number of people, and $15k per month is the conventional total cost (including benefits and even office space) per person. $15k per month is high, so don't actually spend that much. but it's ok to use a high estimate when fundraising to add a margin for error. if you have additional expenses, like manufacturing, add in those at the end. assuming you have none and you think you might hire 20 people, the most you'd want to raise is 20 x $15k x 18 = $5.4 million. [18]

**underestimate how much you want.**

though you can focus on different plans when talking to different types of investors, you should on the whole err on the side of underestimating the amount you hope to raise.

for example, if you'd like to raise $500k, it's better to say initially that you're trying to raise $250k. then when you reach $150k you're more than half done. that sends two useful signals to investors: that you're doing well, and that they have to decide quickly because you're running out of room. whereas if

you'd said you were raising $500k, you'd be less than a third done at $150k. if fundraising stalled there for an appreciable time, you'd start to read as a failure.

saying initially that you're raising $250k doesn't limit you to raising that much. when you reach your initial target and you still have investor interest, you can just decide to raise more. startups do that all the time. in fact, most startups that are very successful at fundraising end up raising more than they originally intended.

i'm not saying you should lie, but that you should lower your expectations initially. there is almost no downside in starting with a low number. it not only won't cap the amount you raise, but will on the whole tend to increase it.

a good metaphor here is angle of attack. if you try to fly at too steep an angle of attack, you just stall. if you say right out of the gate that you want to raise a $5 million series a round, unless you're in a very strong position, you not only won't get that but won't get anything. better to start at a low angle of attack, build up speed, and then gradually increase the angle if you want.

**be profitable if you can.**

you will be in a much stronger position if your collection of plans includes one for raising zero dollars — i.e. if you can make it to profitability without raising any additional money. ideally you want to be able to say to investors "we'll succeed no matter what, but raising money will help us do it faster."

there are many analogies between fundraising and dating, and this is one of the strongest. no one wants you if you seem desperate. and the best way not to seem desperate is not to *be* desperate. that's one reason we urge startups during yc to keep expenses low and to try to make it to ramen profitability before demo day. though it sounds slightly paradoxical, if you want to raise money, the best thing you can do is get yourself to the point where you don't need to.

there are almost two distinct modes of fundraising: one in which founders who need money knock on doors seeking it, knowing that otherwise the company will die or at the very least people will have to be fired, and one in which founders who don't need money take some to grow faster than they could merely on their own revenues. to emphasize the distinction i'm going to name them: type a fundraising is when you don't need money, and type b fundraising is when you do.

inexperienced founders read about famous startups doing what was type a fundraising, and decide they should raise money too, since that seems to be how startups work. except when they raise money they don't have a clear path to profitability and are thus doing type b fundraising. and they are then surprised how difficult and unpleasant it is.

of course not all startups can make it to ramen profitability in a few months. and some that don't still manage to have the upper hand over investors, if they have some other advantage like extraordinary growth numbers or exceptionally formidable founders. but as time passes it gets increasingly difficult to fundraise from a position of strength without being profitable. [19]

**don't optimize for valuation.**

when you raise money, what should your valuation be? the most important thing to understand about valuation is that it's not that important.

founders who raise money at high valuations tend to be unduly proud of it. founders are often competitive people, and since valuation is usually the only visible number attached to a startup, they end up competing to raise money at the highest valuation. this is stupid, because fundraising is not the test

that matters. the real test is revenue. fundraising is just a means to that end. being proud of how well you did at fundraising is like being proud of your college grades.

not only is fundraising not the test that matters, valuation is not even the thing to optimize about fundraising. the number one thing you want from phase 2 fundraising is to get the money you need, so you can get back to focusing on the real test, the success of your company. number two is good investors. valuation is at best third.

the empirical evidence shows just how unimportant it is. dropbox and airbnb are the most successful companies we've funded so far, and they raised money after y combinator at premoney valuations of $4 million and $2.6 million respectively. prices are so much higher now that if you can raise money at all you'll probably raise it at higher valuations than dropbox and airbnb. so let that satisfy your competitiveness. you're doing better than dropbox and airbnb! at a test that doesn't matter.

when you start fundraising, your initial valuation (or valuation cap) will be set by the deal you make with the first investor who commits. you can increase the price for later investors, if you get a lot of interest, but by default the valuation you got from the first investor becomes your asking price.

so if you're raising money from multiple investors, as most companies do in phase 2, you have to be careful to avoid raising the first from an over-eager investor at a price you won't be able to sustain. you can of course lower your price if you need to (in which case you should give the same terms to investors who invested earlier at a higher price), but you may lose a bunch of leads in the process of realizing you need to do this.

what you can do if you have eager first investors is raise money from them on an uncapped convertible note with an mfn clause. this is essentially a way of saying that the valuation cap of the note will be determined by the next investors you raise money from.

it will be easier to raise money at a lower valuation. it shouldn't be, but it is. since phase 2 prices vary at most 10x and the big successes generate returns of at least 100x, investors should pick startups entirely based on their estimate of the probability that the company will be a big success and hardly at all on price. but although it's a mistake for investors to care about price, a significant number do. a startup that investors seem to like but won't invest in at a cap of $x will have an easier time at $x/2. [20]

**yes/no before valuation.**

some investors want to know what your valuation is before they even talk to you about investing. if your valuation has already been set by a prior investment at a specific valuation or cap, you can tell them that number. but if it isn't set because you haven't closed anyone yet, and they try to push you to name a price, resist doing so. if this would be the first investor you've closed, then this could be the tipping point of fundraising. that means closing this investor is the first priority, and you need to get the conversation onto that instead of being dragged sideways into a discussion of price.

fortunately there is a way to avoid naming a price in this situation. and it is not just a negotiating trick; it's how you (both) should be operating. tell them that valuation is not the most important thing to you and that you haven't thought much about it, that you are looking for investors you want to partner with and who want to partner with you, and that you should talk first about whether they want to invest at all. then if they decide they do want to invest, you can figure out a price. but first things first.

since valuation isn't that important and getting fundraising rolling is, we usually tell founders to give the first investor who commits as low a price as they need to. this is a safe technique so long as you combine it with the next one. [21]

**beware "valuation sensitive" investors.**

occasionally you'll encounter investors who describe themselves as "valuation sensitive." what this means in practice is that they are compulsive negotiators who will suck up a lot of your time trying to push your price down. you should therefore never approach such investors first. while you shouldn't chase high valuations, you also don't want your valuation to be set artificially low because the first investor who committed happened to be a compulsive negotiator. some such investors have value, but the time to approach them is near the end of fundraising, when you're in a position to say "this is the price everyone else has paid; take it or leave it" and not mind if they leave it. this way, you'll not only get market price, but it will also take less time.

ideally you know which investors have a reputation for being "valuation sensitive" and can postpone dealing with them till last, but occasionally one you didn't know about will pop up early on. the rule of doing breadth first search weighted by expected value already tells you what to do in this case: slow down your interactions with them.

there are a handful of investors who will try to invest at a lower valuation even when your price has already been set. lowering your price is a backup plan you resort to when you discover you've let the price get set too high to close all the money you need. so you'd only want to talk to this sort of investor if you were about to do that anyway. but since investor meetings have to be arranged at least a few days in advance and you can't predict when you'll need to resort to lowering your price, this means in practice that you should approach this type of investor last if at all.

if you're surprised by a lowball offer, treat it as a backup offer and delay responding to it. when someone makes an offer in good faith, you have a moral obligation to respond in a reasonable time. but lowballing you is a dick move that should be met with the corresponding countermove.

**accept offers greedily.**

i'm a little leery of using the term "greedily" when writing about fundraising lest non-programmers misunderstand me, but a greedy algorithm is simply one that doesn't try to look into the future. a greedy algorithm takes the best of the options in front of it right now. and that is how startups should approach fundraising in phases 2 and later. don't try to look into the future because (a) the future is unpredictable, and indeed in this business you're often being deliberately misled about it and (b) your first priority in fundraising should be to get it finished and get back to work anyway.

if someone makes you an acceptable offer, take it. if you have multiple incompatible offers, take the best. don't reject an acceptable offer in the hope of getting a better one in the future.

these simple rules cover a wide variety of cases. if you're raising money from many investors, roll them up as they say yes. as you start to feel you've raised enough, the threshold for acceptable will start to get higher.

in practice offers exist for stretches of time, not points. so when you get an acceptable offer that would be incompatible with others (e.g. an offer to invest most of the money you need), you can tell the other investors you're talking to that you have an offer good enough to accept, and give them a few days to make their own. this could lose you some that might have made an offer if they had more time. but by definition you don't care; the initial offer was acceptable.

some investors will try to prevent others from having time to decide by giving you an "exploding" offer, meaning one that's only valid for a few days. offers from the very best investors explode less frequently and less rapidly — fred wilson never gives exploding offers, for example — because they're confident

you'll pick them. but lower-tier investors sometimes give offers with very short fuses, because they believe no one who had other options would choose them. a deadline of three working days is acceptable. you shouldn't need more than that if you've been talking to investors in parallel. but a deadline any shorter is a sign you're dealing with a sketchy investor. you can usually call their bluff, and you may need to. [22]

it might seem that instead of accepting offers greedily, your goal should be to get the best investors as partners. that is certainly a good goal, but in phase 2 "get the best investors" only rarely conflicts with "accept offers greedily," because the best investors don't usually take any longer to decide than the others. the only case where the two strategies give conflicting advice is when you have to forgo an offer from an acceptable investor to see if you'll get an offer from a better one. if you talk to investors in parallel and push back on exploding offers with excessively short deadlines, that will almost never happen. but if it does, "get the best investors" is in the average case bad advice. the best investors are also the most selective, because they get their pick of all the startups. they reject nearly everyone they talk to, which means in the average case it's a bad trade to exchange a definite offer from an acceptable investor for a potential offer from a better one.

(the situation is different in phase 1. you can't apply to all the incubators in parallel, because some offset their schedules to prevent this. in phase 1, "accept offers greedily" and "get the best investors" do conflict, so if you want to apply to multiple incubators, you should do it in such a way that the ones you want most decide first.)

sometimes when you're raising money from multiple investors, a series a will emerge out of those conversations, and these rules even cover what to do in that case. when an investor starts to talk to you about a series a, keep taking smaller investments till they actually give you a termsheet. there's no practical difficulty. if the smaller investments are on convertible notes, they'll just convert into the series a round. the series a investor won't like having all these other random investors as bedfellows, but if it bothers them so much they should get on with giving you a termsheet. till they do, you don't know for sure they will, and the greedy algorithm tells you what to do. [23]

**don't sell more than 25% in phase 2.**

if you do well, you will probably raise a series a round eventually. i say probably because things are changing with series a rounds. startups may start to skip them. but only one company we've funded has so far, so tentatively assume the path to huge passes through an a round. [24]

which means you should avoid doing things in earlier rounds that will mess up raising an a round. for example, if you've sold more than about 40% of your company total, it starts to get harder to raise an a round, because vcs worry there will not be enough stock left to keep the founders motivated.

our rule of thumb is not to sell more than 25% in phase 2, on top of whatever you sold in phase 1, which should be less than 15%. if you're raising money on uncapped notes, you'll have to guess what the eventual equity round valuation might be. guess conservatively.

(since the goal of this rule is to avoid messing up the series a, there's obviously an exception if you end up raising a series a in phase 2, as a handful of startups do.)

**have one person handle fundraising.**

if you have multiple founders, pick one to handle fundraising so the other(s) can keep working on the company. and since the danger of fundraising is not the time taken up by the actual meetings but that

it becomes the top idea in your mind, the founder who handles fundraising should make a conscious effort to insulate the other founder(s) from the details of the process. [25]

(if the founders mistrust one another, this could cause some friction. but if the founders mistrust one another, you have worse problems to worry about than how to organize fundraising.)

the founder who handles fundraising should be the ceo, who should in turn be the most formidable of the founders. even if the ceo is a programmer and another founder is a salesperson? yes. if you happen to be that type of founding team, you're effectively a single founder when it comes to fundraising.

it's ok to bring all the founders to meet an investor who will invest a lot, and who needs this meeting as the final step before deciding. but wait till that point. introducing an investor to your cofounder(s) should be like introducing a girl/boyfriend to your parents — something you do only when things reach a certain stage of seriousness.

even if there are still one or more founders focusing on the company during fundraising, growth will slow. but try to get as much growth as you can, because fundraising is a segment of time, not a point, and what happens to the company during that time affects the outcome. if your numbers grow significantly between two investor meetings, investors will be hot to close, and if your numbers are flat or down they'll start to get cold feet.

**you'll need an executive summary and (maybe) a deck.**

traditionally phase 2 fundraising consists of presenting a slide deck in person to investors. sequoia describes what such a deck should contain, and since they're the customer you can take their word for it.

i say "traditionally" because i'm ambivalent about decks, and (though perhaps this is wishful thinking) they seem to be on the way out. a lot of the most successful startups we fund never make decks in phase 2. they just talk to investors and explain what they plan to do. fundraising usually takes off fast for the startups that are most successful at it, and they're thus able to excuse themselves by saying that they haven't had time to make a deck.

you'll also want an executive summary, which should be no more than a page long and describe in the most matter of fact language what you plan to do, why it's a good idea, and what progress you've made so far. the point of the summary is to remind the investor (who may have met many startups that day) what you talked about.

assume that if you give someone a copy of your deck or executive summary, it will be passed on to whoever you'd least like to have it. but don't refuse on that account to give copies to investors you meet. you just have to treat such leaks as a cost of doing business. in practice it's not that high a cost. though founders are rightly indignant when their plans get leaked to competitors, i can't think of a startup whose outcome has been affected by it.

sometimes an investor will ask you to send them your deck and/or executive summary before they decide whether to meet with you. i wouldn't do that. it's a sign they're not really interested.

**stop fundraising when it stops working.**

when do you stop fundraising? ideally when you've raised enough. but what if you haven't raised as much as you'd like? when do you give up?

it's hard to give general advice about this, because there have been cases of startups that kept trying to raise money even when it seemed hopeless, and miraculously succeeded. but what i usually tell founders is to stop fundraising when you start to get a lot of air in the straw. when you're drinking

through a straw, you can tell when you get to the end of the liquid because you start to get a lot of air in the straw. when your fundraising options run out, they usually run out in the same way. don't keep sucking on the straw if you're just getting air. it's not going to get better.

**don't get addicted to fundraising.**

fundraising is a chore for most founders, but some find it more interesting than working on their start-up. the work at an early stage startup often consists of unglamorous schleps. whereas fundraising, when it's going well, can be quite the opposite. instead of sitting in your grubby apartment listening to users complain about bugs in your software, you're being offered millions of dollars by famous investors over lunch at a nice restaurant. [26]

the danger of fundraising is particularly acute for people who are good at it. it's always fun to work on something you're good at. if you're one of these people, beware. fundraising is not what will make your company successful. listening to users complain about bugs in your software is what will make you successful. and the big danger of getting addicted to fundraising is not merely that you'll spend too long on it or raise too much money. it's that you'll start to think of yourself as being already successful, and lose your taste for the schleps you need to undertake to actually be successful. startups can be destroyed by this.

when i see a startup with young founders that is fabulously successful at fundraising, i mentally decrease my estimate of the probability that they'll succeed. the press may be writing about them as if they'd been anointed as the next google, but i'm thinking "this is going to end badly."

**don't raise too much.**

though only a handful of startups have to worry about this, it is possible to raise too much. the dangers of raising too much are subtle but insidious. one is that it will set impossibly high expectations. if you raise an excessive amount of money, it will be at a high valuation, and the danger of raising money at too high a valuation is that you won't be able to increase it sufficiently the next time you raise money.

a company's valuation is expected to rise each time it raises money. if not it's a sign of a company in trouble, which makes you unattractive to investors. so if you raise money in phase 2 at a post-money valuation of $30 million, the pre-money valuation of your next round, if you want to raise one, is going to have to be at least $50 million. and you have to be doing really, really well to raise money at $50 million.

it's very dangerous to let the competitiveness of your current round set the performance threshold you have to meet to raise your next one, because the two are only loosely coupled.

but the money itself may be more dangerous than the valuation. the more you raise, the more you spend, and spending a lot of money can be disastrous for an early stage startup. spending a lot makes it harder to become profitable, and perhaps even worse, it makes you more rigid, because the main way to spend money is people, and the more people you have, the harder it is to change directions. so if you do raise a huge amount of money, don't spend it. (you will find that advice almost impossible to follow, so hot will be the money burning a hole in your pocket, but i feel obliged at least to try.)

**be nice.**

startups raising money occasionally alienate investors by seeming arrogant. sometimes because they are arrogant, and sometimes because they're noobs clumsily attempting to mimic the toughness they've observed in experienced founders.

it's a mistake to behave arrogantly to investors. while there are certain situations in which certain investors like certain kinds of arrogance, investors vary greatly in this respect, and a flick of the whip that will bring one to heel will make another roar with indignation. the only safe strategy is never to seem arrogant at all.

that will require some diplomacy if you follow the advice i've given here, because the advice i've given is essentially how to play hardball back. when you refuse to meet an investor because you're not in fundraising mode, or slow down your interactions with an investor who moves too slow, or treat a contingent offer as the no it actually is and then, by accepting offers greedily, end up leaving that investor out, you're going to be doing things investors don't like. so you must cushion the blow with soft words. at yc we tell startups they can blame us. and now that i've written this, everyone else can blame me if they want. that plus the inexperience card should work in most situations: sorry, we think you're great, but pg said startups shouldn't ___, and since we're new to fundraising, we feel like we have to play it safe.

the danger of behaving arrogantly is greatest when you're doing well. when everyone wants you, it's hard not to let it go to your head. especially if till recently no one wanted you. but restrain yourself. the startup world is a small place, and startups have lots of ups and downs. this is a domain where it's more true than usual that pride goeth before a fall. [27]

be nice when investors reject you as well. the best investors are not wedded to their initial opinion of you. if they reject you in phase 2 and you end up doing well, they'll often invest in phase 3. in fact investors who reject you are some of your warmest leads for future fundraising. any investor who spent significant time deciding probably came close to saying yes. often you have some internal champion who only needs a little more evidence to convince the skeptics. so it's wise not merely to be nice to investors who reject you, but (unless they behaved badly) to treat it as the beginning of a relationship.

**the bar will be higher next time.**

assume the money you raise in phase 2 will be the last you ever raise. you must make it to profitability on this money if you can.

over the past several years, the investment community has evolved from a strategy of anointing a small number of winners early and then supporting them for years to a strategy of spraying money at early stage startups and then ruthlessly culling them at the next stage. this is probably the optimal strategy for investors. it's too hard to pick winners early on. better to let the market do it for you. but it often comes as a surprise to startups how much harder it is to raise money in phase 3.

when your company is only a couple months old, all it has to be is a promising experiment that's worth funding to see how it turns out. the next time you raise money, the experiment has to have worked. you have to be on a trajectory that leads to going public. and while there are some ideas where the proof that the experiment worked might consist of e.g. query response times, usually the proof is profitability. usually phase 3 fundraising has to be type a fundraising.

in practice there are two ways startups hose themselves between phases 2 and 3. some are just too slow to become profitable. they raise enough money to last for two years. there doesn't seem any particular urgency to be profitable. so they don't make any effort to make money for a year. but by that time, not making money has become habitual. when they finally decide to try, they find they can't.

the other way companies hose themselves is by letting their expenses grow too fast. which almost always means hiring too many people. you usually shouldn't go out and hire 8 people as soon as you raise money at phase 2. usually you want to wait till you have growth (and thus usually revenues) to

justify them. a lot of vcs will encourage you to hire aggressively. vcs generally tell you to spend too much, partly because as money people they err on the side of solving problems by spending money, and partly because they want you to sell them more of your company in subsequent rounds. don't listen to them.

**don't make things complicated.**

i realize it may seem odd to sum up this huge treatise by saying that my overall advice is not to make fundraising too complicated, but if you go back and look at this list you'll see it's basically a simple recipe with a lot of implications and edge cases. avoid investors till you decide to raise money, and then when you do, talk to them all in parallel, prioritized by expected value, and accept offers greedily. that's fundraising in one sentence. don't introduce complicated optimizations, and don't let investors introduce complications either.

fundraising is not what will make you successful. it's just a means to an end. your primary goal should be to get it over with and get back to what will make you successful — making things and talking to users — and the path i've described will for most startups be the surest way to that destination.

be good, take care of yourselves, and *don't leave the path*.

---

## notes

1 the worst explosions happen when unpromising-seeming startups encounter mediocre investors. good investors don't lead startups on; their reputations are too valuable. and startups that seem promising can usually get enough money from good investors that they don't have to talk to mediocre ones. it is the unpromising-seeming startups that have to resort to raising money from mediocre investors. and it's particularly damaging when these investors flake, because unpromising-seeming startups are usually more desperate for money.

(not all unpromising-seeming startups do badly. some are merely ugly ducklings in the sense that they violate current startup fashions.)

2 one yc founder told me:

> i think in general we've done ok at fundraising, but i managed to screw up twice at the exact same thing — trying to focus on building the company and fundraising at the same time.

3 there is one subtle danger you have to watch out for here, which i warn about later: beware of getting too high a valuation from an eager investor, lest that set an impossibly high target when raising additional money.

4 if they really need a meeting, then they're not ready to invest, regardless of what they say. they're still deciding, which means you're being asked to come in and convince them. which is fundraising.

5 associates at vc firms regularly cold email startups. naive founders think "wow, a vc is interested in us!" but an associate is not a vc. they have no decision-making power. and while they may introduce startups they like to partners at their firm, the partners discriminate against deals that come to them this way. i don't know of a single vc investment that began with an associate cold-emailing a startup. if you want to approach a specific firm, get an intro to a partner from someone they respect.

it's ok to talk to an associate if you get an intro to a vc firm or they see you at a demo day and they begin by having an associate vet you. that's not a promising lead and should therefore get low priority, but it's not as completely worthless as a cold email.

because the title "associate" has gotten a bad reputation, a few vc firms have started to give their associates the title "partner," which can make things very confusing. if you're a yc startup you can ask us who's who; otherwise you may have to do some research online. there may be a special title for actual partners. if someone speaks for the firm in the press or a blog on the firm's site, they're probably a real partner. if they're on boards of directors they're probably a real partner.

there are titles between "associate" and "partner," including "principal" and "venture partner." the meanings of these titles vary too much to generalize.

6 for similar reasons, avoid casual conversations with potential acquirers. they can lead to distractions even more dangerous than fundraising. don't even take a meeting with a potential acquirer unless you want to sell your company right now.

7 joshua reeves specifically suggests asking each investor to intro you to two more investors.

don't ask investors who say no for introductions to other investors. that will in many cases be an anti-recommendation.

8 this is not always as deliberate as its sounds. a lot of the delays and disconnects between founders and investors are induced by the customs of the venture business, which have evolved the way they have because they suit investors' interests.

9 one yc founder who read a draft of this essay wrote:

> this is the most important section. i think it might bear stating even more clearly. "investors will deliberately affect more interest than they have to preserve optionality. if an investor seems very interested in you, they still probably won't invest. the solution for this is to assume the worst — that an investor is just feigning interest — until you get a definite commitment."

10 though you should probably pack investor meetings as closely as you can, jeff byun mentions one reason not to: if you pack investor meetings too closely, you'll have less time for your pitch to evolve.

some founders deliberately schedule a handful of lame investors first, to get the bugs out of their pitch.

11 there is not an efficient market in this respect. some of the most useless investors are also the highest maintenance.

12 incidentally, this paragraph is sales 101. if you want to see it in action, go talk to a car dealer.

13 i know one very smooth founder who used to end investor meetings with "so, can i count you in?" delivered as if it were "can you pass the salt?" unless you're very smooth (if you're not sure...), do not do this yourself. there is nothing more unconvincing, for an investor, than a nerdy founder trying to deliver the lines meant for a smooth one.

investors are fine with funding nerds. so if you're a nerd, just try to be a good nerd, rather than doing a bad imitation of a smooth salesman.

14 ian hogarth suggests a good way to tell how serious potential investors are: the resources they expend on you after the first meeting. an investor who's seriously interested will already be working to help you even before they've committed.

15 in principle you might have to think about so-called "signalling risk." if a prestigious vc makes a small seed investment in you, what if they don't want to invest the next time you raise money? other investors might assume that the vc knows you well, since they're an existing investor, and if they don't want to invest in your next round, that must mean you suck. the reason i say "in principle" is that in practice signalling hasn't been much of a problem so far. it rarely arises, and in the few cases where it does, the startup in question usually is doing badly and is doomed anyway.

if you have the luxury of choosing among seed investors, you can play it safe by excluding vc firms. but it isn't critical to.

16 sometimes a competitor will deliberately threaten you with a lawsuit just as you start fundraising, because they know you'll have to disclose the threat to potential investors and they hope this will make it harder for you to raise money. if this happens it will probably frighten you more than investors. experienced investors know about this trick, and know the actual lawsuits rarely happen. so if you're attacked in this way, be forthright with investors. they'll be more alarmed if you seem evasive than if you tell them everything.

17 a related trick is to claim that they'll only invest contingently on other investors doing so because otherwise you'd be "undercapitalized." this is almost always bullshit. they can't estimate your minimum capital needs that precisely.

18 you won't hire all those 20 people at once, and you'll probably have some revenues before 18 months are out. but those too are acceptable or at least accepted additions to the margin for error.

19 type a fundraising is so much better that it might even be worth doing something different if it gets you there sooner. one yc founder told me that if he were a first-time founder again he'd "leave ideas that are up-front capital intensive to founders with established reputations."

20 i don't know whether this happens because they're innumerate, or because they believe they have zero ability to predict startup outcomes (in which case this behavior at least wouldn't be irrational). in either case the implications are similar.

21 if you're a yc startup and you have an investor who for some reason insists that you decide the price, any yc partner can estimate a market price for you.

22 you should respond in kind when investors behave upstandingly too. when an investor makes you a clean offer with no deadline, you have a moral obligation to respond promptly.

23 tell the investors talking to you about an a round about the smaller investments you raise as you raise them. you owe them such updates on your cap table, and this is also a good way to pressure them to act. they won't like you raising other money and may pressure you to stop, but they can't legitimately ask you to commit to them till they also commit to you. if they want you to stop raising money, the way to do it is to give you a series a termsheet with a no-shop clause.

you can relent a little if the potential series a investor has a great reputation and they're clearly working fast to get you a termsheet, particularly if a third party like yc is involved to ensure there are no misunderstandings. but be careful.

24 the company is weebly, which made it to profitability on a seed investment of $650k. they did try to raise a series a in the fall of 2008 but (no doubt partly because it was the fall of 2008) the terms they were offered were so bad that they decided to skip raising an a round.

25 another advantage of having one founder take fundraising meetings is that you never have to negotiate in real time, which is something inexperienced founders should avoid. one yc founder told me:

> investors are professional negotiators and can negotiate on the spot very easily. if only one founder is in the room, you can say "i need to circle back with my co-founder" before making any commitments. i used to do this all the time.

26 you'll be lucky if fundraising feels pleasant enough to become addictive. more often you have to worry about the other extreme — becoming demoralized when investors reject you. as one (very successful) yc founder wrote after reading a draft of this:

> it's hard to mentally deal with the sheer scale of rejection in fundraising and if you are not in the right mindset you will fail. users may love you but these supposedly smart investors may not understand you at all. at this point for me, rejection still rankles but i've come to accept that investors are just not super thoughtful for the most part and you need to play the game according to certain somewhat depressing rules (many of which you are listing) in order to win.

27 the actual sentence in the king james bible is "pride goeth before destruction, and an haughty spirit before a fall."

**thanks** to slava akhmechet, sam altman, nate blecharczyk, adora cheung, bill clerico, john collison, patrick collison, parker conrad, ron conway, travis deyle, jason freedman, joe gebbia, mattan griffel, kevin hale, jacob heller, ian hogarth, justin kan, professor moriarty, nikhil nirmel, david petersen, geoff ralston, joshua reeves, yuri sagalov, emmett shear, rajat suri, garry tan, and nick tomarello for reading drafts of this.

12

# donate unrestricted

march 2021
paulgraham.com/donate.html

the secret curse of the nonprofit world is restricted donations. if you haven't been involved with nonprofits, you may never have heard this phrase before. but if you have been, it probably made you wince.

restricted donations mean donations where the donor limits what can be done with the money. this is common with big donations, perhaps the default. and yet it's usually a bad idea. usually the way the donor wants the money spent is not the way the nonprofit would have chosen. otherwise there would have been no need to restrict the donation. but who has a better understanding of where money needs to be spent, the nonprofit or the donor?

if a nonprofit doesn't understand better than its donors where money needs to be spent, then it's incompetent and you shouldn't be donating to it at all.

which means a restricted donation is inherently suboptimal. it's either a donation to a bad nonprofit, or a donation for the wrong things.

there are a couple exceptions to this principle. one is when the nonprofit is an umbrella organization. it's reasonable to make a restricted donation to a university, for example, because a university is only nominally a single nonprofit. another exception is when the donor actually does know as much as the nonprofit about where money needs to be spent. the gates foundation, for example, has specific goals and often makes restricted donations to individual nonprofits to accomplish them. but unless you're a domain expert yourself or donating to an umbrella organization, your donation would do more good if it were unrestricted.

if restricted donations do less good than unrestricted ones, why do donors so often make them? partly because doing good isn't donors' only motive. they often have other motives as well — to make a mark, or to generate good publicity [1], or to comply with regulations or corporate policies. many donors may simply never have considered the distinction between restricted and unrestricted donations. they may believe that donating money for some specific purpose is just how donation works. and to be fair, nonprofits don't try very hard to discourage such illusions. they can't afford to. people running nonprofits are almost always anxious about money. they can't afford to talk back to big donors.

you can't expect candor in a relationship so asymmetric. so i'll tell you what nonprofits wish they could tell you. if you want to donate to a nonprofit, donate unrestricted. if you trust them to spend your money, trust them to decide how.

---

## notes

1 unfortunately restricted donations tend to generate more publicity than unrestricted ones. "x donates money to build a school in africa" is not only more interesting than "x donates money to y nonprofit to spend as y chooses," but also focuses more attention on x.

**thanks** to chase adam, ingrid bassett, trevor blackwell, and edith elliot for reading drafts of this.

01 how to make wealth
02 how people get rich now
03 billionaires build
04 ramen profitable
05 default alive or default dead?
06 the fatal pinch
07 the equity equation
08 investor herd dynamics
09 high resolution fundraising
10 how to convince investors
11 how to raise money
12 donate unrestricted


12 essays · 37,632 words · paulgraham.com